MECHATRONICS AND MEMS

Subject Code : 23A03702d

UNIT I : Introduction

Syllabus: Definition of Mechatronics, Need for Mechatronics in Industry, Objectives of mechatronics, mechatronics design process, Mechatronics key elements, applications - CNC machines, Tool monitoring systems, FMS, Industrial Robots, Automatic packaging systems, Automatic inspection systems.

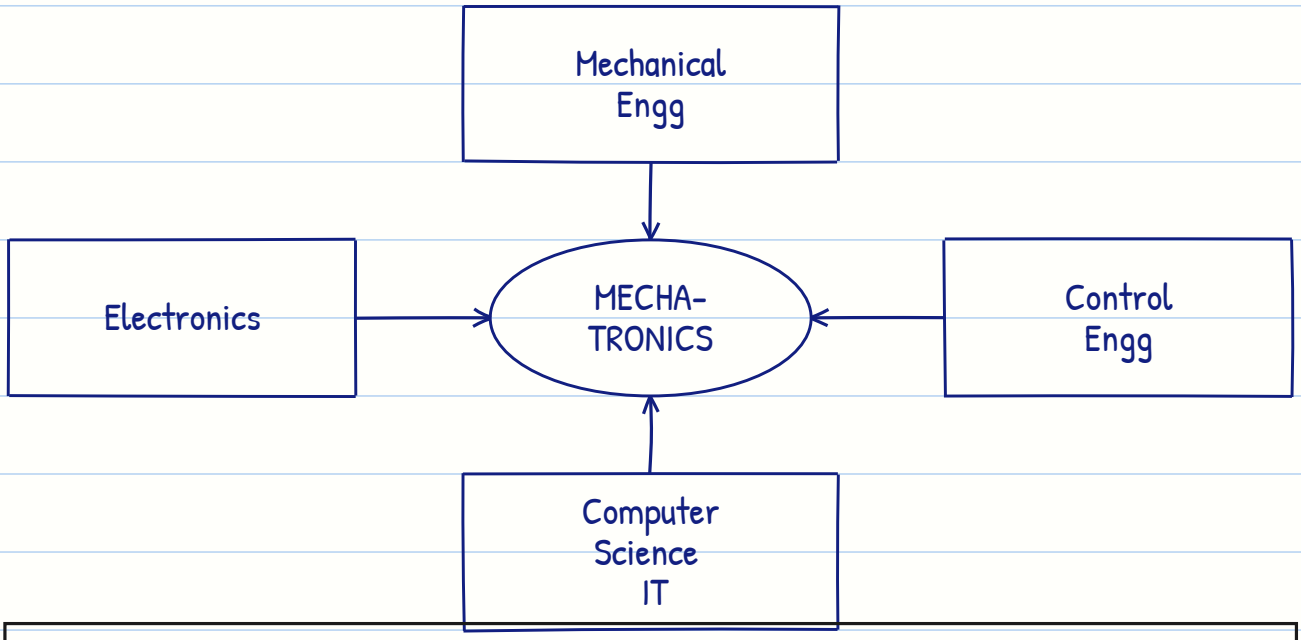
I. Definition of Mechatronics

The word **Mechatronics** was coined in 1969 by **Tetsuro Mori** , an engineer at Yaskawa Electric Corporation, Japan. It is made from two parts: **MECHA** from Mechanics (mechanical engineering) and **TRONICS** from Electronics.

Definitions

- Mechatronics is the **synergistic integration** of mechanical engineering, electronics, control engineering and computer science in the design and manufacture of products and processes.
- IEEE/ASME: "Synergistic integration of mechanical engineering with electronics and intelligent computer control in the design and manufacturing of industrial products and processes."
- Mechatronics = Mechanical + Electronics + Electrical + Control + Computer (IT) working together as ONE system.

- A mechatronic system is **intelligent** : it senses the environment, processes the information and acts on it automatically.



Mechatronics = Mechanics + Electronics + Control + Computing

Examples of mechatronic products

- Washing machine, microwave oven, digital camera, ATM, printers, CD/DVD drive, ABS braking and airbags in cars, CNC machine, robots, drones, 3D printer, automatic door, smart phone.

Traditional vs Mechatronic approach

Point	Traditional design	Mechatronic design
Approach	Sequential - mech first, then electrical	Concurrent - all domains together
Flexibility	Fixed by mechanical parts (cams, gears)	Changed by software
Accuracy	Limited, wear affects it	High, feedback corrects errors
Size & cost	Bulky, costly	Compact, cheaper

Intelligence	None or little	Self-monitoring, decision making

2. Need for Mechatronics in Industry

- **Global competition:** products must be cheap, good quality and reach the market fast (short time to market).
- **Customer demand:** customised products, small batch sizes, variety. Needs flexible machines.
- **High precision & repeatability:** micron-level accuracy which humans cannot give.
- **Productivity:** machines work 24 x 7 without fatigue, so output goes up and unit cost goes down.
- **Quality:** feedback sensors and automatic inspection reduce rejects (zero-defect aim).
- **Safety:** robots do dangerous jobs - painting, welding, handling hot or toxic materials.
- **Compact and light designs:** electronics replaces heavy mechanical linkages, cams and gear trains.
- **Reliability & less maintenance:** fewer moving parts, self-diagnosis and predictive maintenance.
- **Flexibility:** change of product only needs change of program - no new tooling.
- **Energy and material saving:** accurate control avoids wastage.

Advantages

- Higher accuracy and speed, lower cost per piece, better quality.
- Flexible, programmable, easy to upgrade (software update).
- Remote monitoring and data collection (Industry 4.0, IoT).

Disadvantages / limitations

- High initial investment and need for skilled multi-disciplinary engineers.
- Complex design, difficult fault finding (a fault may be in mechanics, electronics or software).
- Needs a stable power supply, and is affected by noise and vibration.

3. Objectives of Mechatronics

- To **integrate** mechanical, electronic, control and computer technologies for a better product.
- To build **smart / intelligent** products that adapt to changes in their surroundings.
- To increase **productivity, accuracy and repeatability** of machines.
- To reduce **cost, size, weight and energy consumption** of the product.
- To improve **reliability** and reduce maintenance.
- To get **flexibility** - the same hardware doing many tasks by changing software.
- To reduce **design time** and time-to-market by concurrent engineering.

- To improve **safety** and user friendliness (better man-machine interface).
- To enable **automation** of manufacturing and remote operation.

4. Mechatronics Design Process

Mechatronic design is **concurrent (simultaneous) engineering** : mechanical, electrical and software teams work together from the beginning, not one after another. It follows the **V-model** .

Stages

- **Need / requirement analysis:** identify the customer need and write a clear specification (speed, load, accuracy, cost, size, safety).
- **Conceptual design:** generate alternative solutions, select the best with a decision matrix. Decide what is done in mechanics, in electronics and in software.
- **System-level design:** divide the system into subsystems (sensor, actuator, controller, mechanism) and fix interfaces between them.
- **Detailed design:** mechanical (CAD, stress), electrical (circuits, PCB), software (algorithms, code) and control design (PID tuning).
- **Modelling and simulation:** build mathematical models (MATLAB/Simulink) and check behaviour before making hardware.
- **Prototype & hardware-in-loop test:** integrate and test each module, then the whole system.
- **Validation and optimisation:** check against the specification, tune and improve.

- **Manufacture, documentation and maintenance.**

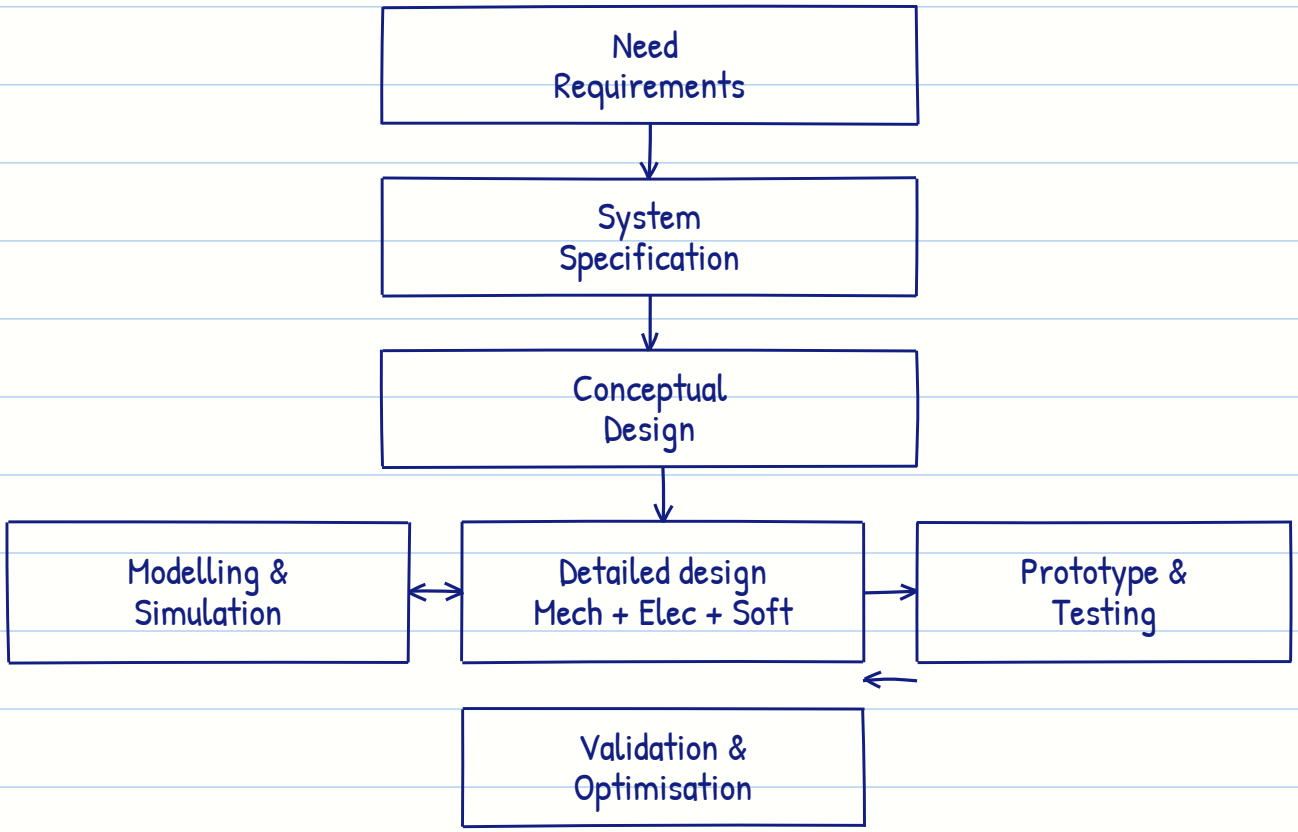


Fig: Mechatronic design process (V-model, concurrent engineering)

Traditional vs Mechatronic design

- Traditional: design is sequential and "thrown over the wall" - errors are found late and costly.
- Mechatronic: parallel design, so the interactions between mechanics, electronics and software are checked early.

5. Key Elements of a Mechatronic System

A mechatronic system has **five key elements** :

- **1. Sensors:** convert a physical quantity (position, force, temperature, speed) into an electrical signal. Eg: encoder, LVDT, thermocouple, strain gauge.

- **2. Signal conditioning:** amplify, filter, convert (ADC/DAC), isolate and linearise the sensor signal so the controller can use it. Eg: op-amp, filters, ADC.
- **3. Controller (decision unit):** compares the actual value with the desired value and takes a decision. Eg: microprocessor, microcontroller, PLC, PC, DSP.
- **4. Actuators (with drivers):** convert the controller's electrical output into physical action - motion, force, heat. Eg: motors, solenoids, hydraulic and pneumatic cylinders, relays, piezo.
- **5. Mechanical system / Physical plant:** the mechanism that does the actual work. Eg: gears, linkages, slides, robot arm.
- Also: **Display / man-machine interface** (keypad, LCD, touch screen) and **Power supply** .

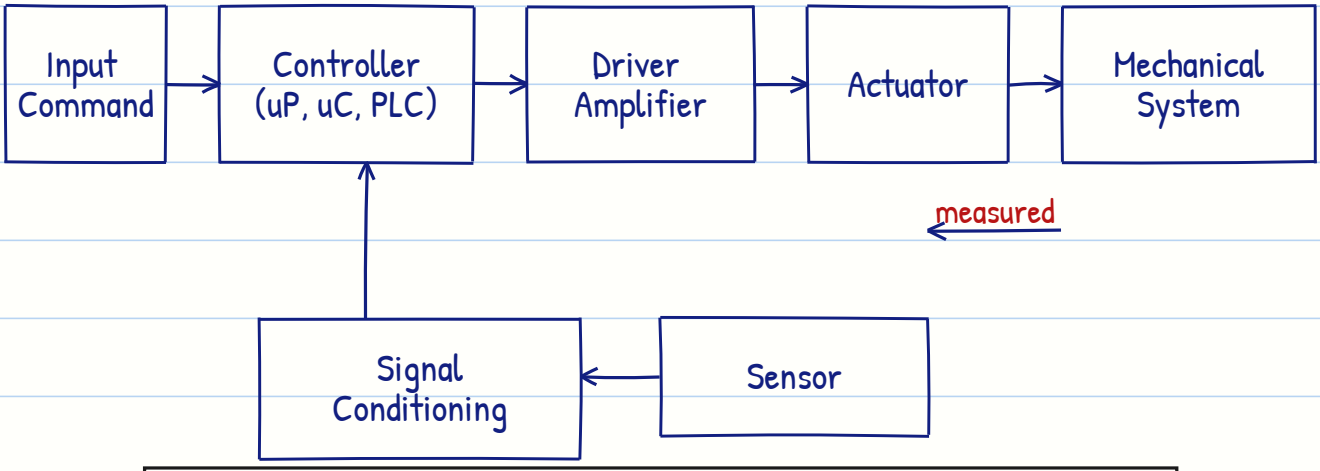


Fig: Closed-loop mechatronic system with feedback

Control loops

- **Open loop:** no feedback; output has no effect on control action. Simple and cheap but not accurate. Eg: toaster, washing machine timer.
- **Closed loop:** the output is measured by a sensor and fed back to the controller to reduce error. Accurate. Eg: CNC servo axis, cruise control, robot joint.
- Error = Set point - Feedback value.

6. Applications of Mechatronics

Mechatronics is used in manufacturing, automobiles, aerospace, medicine, consumer goods, agriculture and defence. Industrial applications in the syllabus are explained below.

6.1 Computer Numerical Control (CNC) Machines

- A CNC machine is a machine tool whose motions (axes, spindle speed, feed, coolant) are controlled by a **computer reading a part program** (G and M codes).
- Eg: CNC lathe, milling machine, drilling machine, machining centre, EDM, laser cutter.

Main parts

- **Part program / input device:** G-code written by the programmer (CAD/CAM) and sent through USB, LAN or keypad.
- **Machine Control Unit (MCU):** the computer. It has a Data

Processing Unit (reads and decodes the program) and a Control Loop Unit (sends signals to drives and takes feedback).

- **Drive system:** servo motor + ball screw for each axis; spindle motor.
- **Feedback devices:** encoders, resolvers and linear scales for position and velocity.
- **Machine tool:** bed, slides, spindle, tool turret/ATC, work holding.
- **Display & operator panel.**

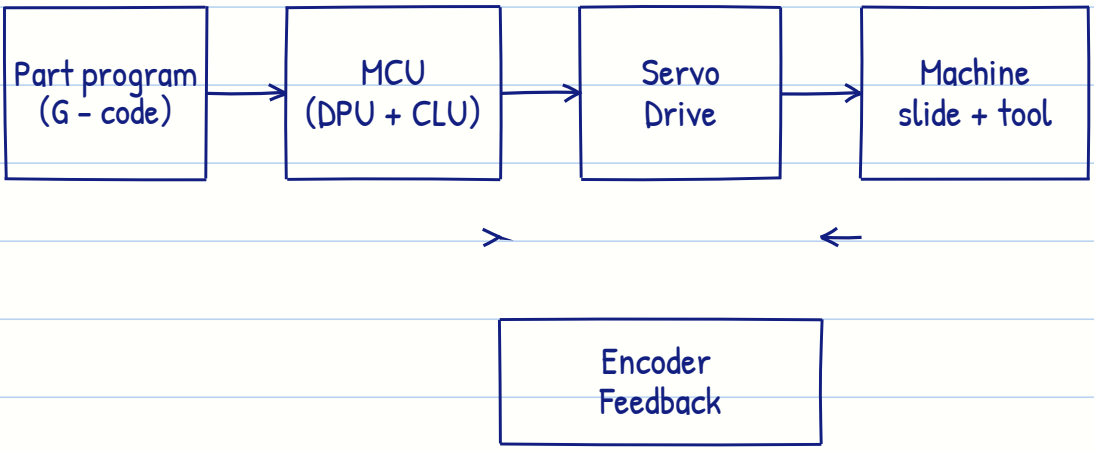


Fig: Block diagram of a CNC machine (closed loop)

Types of control

- **Open loop:** stepper motor, no feedback. Cheap, less accurate.
- **Closed loop:** servo motor with feedback. High accuracy, used in most CNC machines.
- **Point-to-point** (drilling), **straight cut** and **contouring** (continuous path, 2 to 5 axes).

Advantages

- High accuracy and repeatability, good surface finish, complex shapes possible.
- Less setup time and less operator skill, reduced scrap, can run unattended.

Limitations

- Costly machine and maintenance, needs skilled programmer, and is not economic for very small jobs.

6.2 Tool Monitoring Systems

- Tool wear and breakage spoils the job, damages the machine and stops production. A **Tool Condition Monitoring (TCM)** system watches the tool continuously during cutting and gives a warning or stops the machine.
- Tool wear cannot be measured directly during cutting, so it is **indirectly** sensed.

Methods

- **Direct methods:** optical/vision camera, laser, touch probe measure the tool dimension. Done between cuts.
- **Indirect methods:** measure a signal which changes with wear:
  - **Cutting force / torque** (dynamometer, strain gauge) - force rises as the tool wears.
  - **Spindle motor current / power** - increases with wear.
  - **Vibration** (accelerometer) and **acoustic emission (AE)** sensors -

sudden change shows breakage.

- **Temperature** of tool and chip (thermocouple, infrared).
- **Sound** (microphone) and surface roughness of the job.

Working

- Sensors -> signal conditioning (filter, amplifier, ADC) -> feature extraction -> **decision logic** (threshold, neural network, fuzzy) -> alarm or tool change command or machine stop.

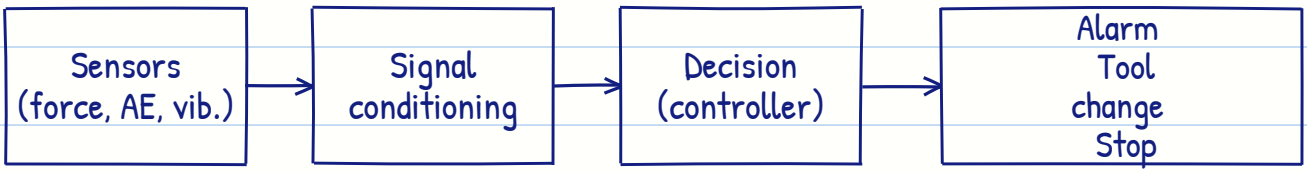


Fig: Tool condition monitoring system

- **Benefits:** avoids scrap and machine crash, increases tool life usage, allows unmanned (lights-out) machining, reduces cost.

6.3 Flexible Manufacturing System (FMS)

- FMS is an **automated manufacturing system** having a group of CNC machines connected by an automated material handling system, all controlled by a central computer, which can make a **variety of parts** with little changeover time.
- It lies between dedicated (transfer line) and stand-alone CNC machines - gives **medium volume, medium variety** .

Main components

- **Workstations:** CNC machines (machining centres, lathes), inspection station (CMM), washing station.
- **Material handling system:** AGVs (automated guided vehicles), conveyors, robots, pallet changers, AS/RS (automated storage & retrieval system).
- **Central computer control:** schedules jobs, sends part programs (DNC), tracks tools and pallets, collects data.
- **Tool management:** tool magazines, tool presetting, automatic tool change.
- **Load/unload stations** and fixtures/pallets.

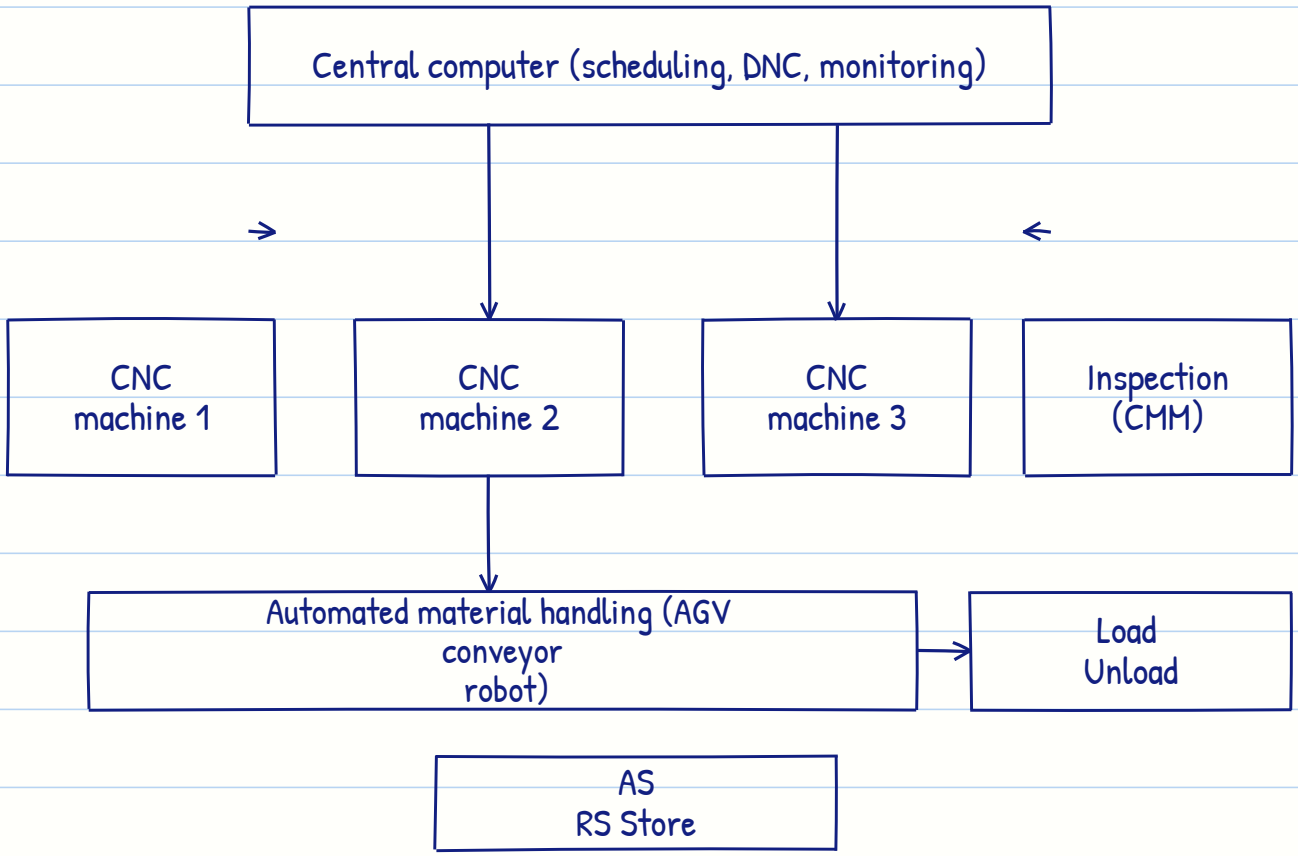


Fig: Layout of a flexible manufacturing system

Types

- **Flexible manufacturing cell (FMC)** - 1 to 3 machines; **FMS** - 4 or more machines; **Flexible transfer line** - high volume.

Advantages

- Reduced setup/changeover and lead time, high machine utilisation (up to 80-90%).
- Low work-in-process inventory, consistent quality, and quick response to design changes.

Disadvantages

- Very high initial cost, complex software, needs skilled people, and the system stops if the central computer fails.

6.4 Industrial Robots

- **Definition (RIA):** A robot is a reprogrammable, multifunctional manipulator designed to move materials, parts, tools or special devices through variable programmed motions for performing a variety of tasks.

Main parts of a robot

- **Manipulator (arm + wrist):** links and joints (revolute R, prismatic P). Gives the degrees of freedom (DOF).
- **End effector:** gripper, welding torch, spray gun, vacuum cup - the "hand" of the robot.
- **Actuators:** electric servo / stepper motors, hydraulic and pneumatic drives.

- **Sensors:** encoders, force/torque, vision, proximity - for feedback.
- **Controller:** the brain, runs program and does kinematics. Teach pendant for programming.
- **Power supply.**

Classification (by configuration)

Type	Joints	Work envelope / use
Cartesian	PPP	Rectangular - pick & place, CNC loading, 3D printing
Cylindrical	RPP	Cylindrical - assembly, machine loading
Spherical (Polar)	RRP	Spherical - die casting, welding
SCARA	RRP (horizontal)	Circular - fast assembly of PCB, small parts
Articulated	RRR	Spherical large - welding, painting

Applications

- Welding, painting, pick and place, assembly, machine tending, palletising, material handling, hazardous work, packaging.

Advantages / limits

- Fast, accurate, tireless, safe for humans, repeatable. Costly, limited sensing/thinking ability, need programming and safety fences.

6.5 Automatic Packaging Systems

- Packaging systems automatically fill, seal, label and pack products

at high speed with the help of sensors, PLCs and actuators. Used in food, pharma, beverages, cement, FMCG.

Operations

- **Feeding / conveying -> Filling (volume or weight) -> Sealing / capping -> Labelling and date printing -> Inspection (weight, leak, vision) -> Carton packing -> Palletising.**

Elements

- **Sensors:** photoelectric (bottle presence), proximity, load cell (weight), level sensor, colour/vision sensor.
- **Actuators:** conveyor motors, pneumatic cylinders (pushers, sealers), solenoid valves, servo for cutters.
- **Controller:** PLC with HMI (touch screen), counters, timers.



Fig: Automatic packaging line

- **Advantages:** high speed, uniform quantity, hygiene, less labour, less wastage, accurate count and weight.

6.6 Automatic Inspection Systems

- An automatic inspection system measures or checks the quality of

every product online (100% inspection) without human involvement and rejects defective parts.

Methods / devices

- **Machine vision:** camera + lighting + image processing software to check dimensions, shape, colour, surface defects, presence of parts, bar codes.
- **Laser scanners / laser micrometers:** non contact measurement of diameter, thickness, profile.
- **Coordinate Measuring Machine (CMM):** touch trigger probe measures 3D coordinates.
- **Proximity / photoelectric and load cells:** presence and weight checks. Also ultrasonic and eddy current for hidden flaws.

Working

- Part arrives on conveyor -> sensor triggers camera -> image captured -> processing and comparison with the standard -> **accept / reject decision** -> reject mechanism (air jet, pusher) removes the faulty part -> data stored for SPC (statistical process control).

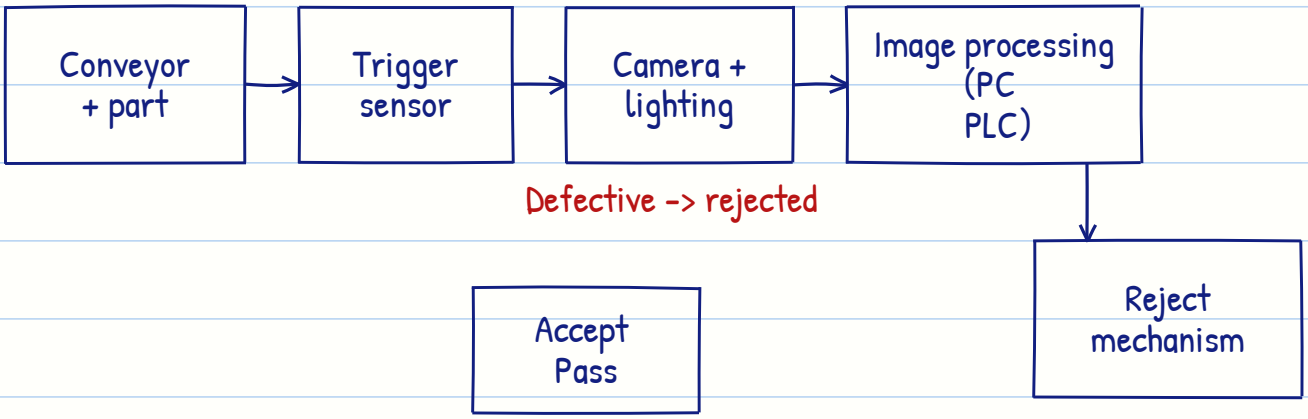


Fig: Automatic vision inspection system

- **Advantages:** 100% inspection, high speed, no fatigue or human error, consistent, data for quality improvement, non contact.

Important Questions (Unit I)

- Define mechatronics. Explain the need and objectives of mechatronics in industry.
- Explain the mechatronics design process with a neat block diagram.
- Explain the key elements of a mechatronic system with an example.
- Describe a CNC machine with a block diagram; write its advantages.
- Write a short note on tool monitoring systems, FMS, industrial robots, automatic packaging and inspection systems.

MECHATRONICS AND MEMS

Subject Code : 23A03702d

UNIT II : Sensors

Syllabus: Static and dynamic characteristics of sensors, Displacement, Position and Proximity sensors, Force and torque sensors, Pressure sensors, Flow sensors, Temperature sensors, Acceleration sensors, Level sensors, Light sensors, Smart material sensors, Micro and Nano sensors, Selection criteria for sensors.

I. Introduction to Sensors

- A **sensor** is a device that detects a physical quantity (measurand) such as displacement, force, temperature, flow, and gives an output signal, usually electrical, proportional to it.
- A **transducer** converts one form of energy to another. Sensor = sensing element; transducer = sensing element + signal conversion. (In practice both words are used for each other.)
- **Classification**
  - **Active sensors** (self generating): give voltage or current without external power. Eg: thermocouple, piezoelectric, photovoltaic.
  - **Passive sensors:** need external excitation; they change R, L or C. Eg: strain gauge, RTD, LVDT, thermistor.
  - **Analog sensors** give continuous output (LVDT, potentiometer); **digital sensors** give pulses or codes (encoder).
  - By measurand: displacement, force, pressure, flow, temperature, level,

light, acceleration etc.

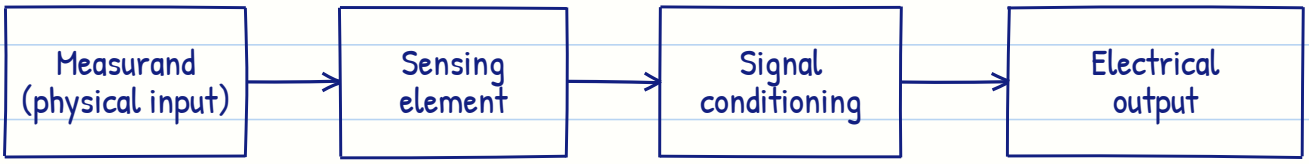


Fig: Sensor - basic block diagram

2. Static Characteristics of Sensors

Static characteristics describe the output when the input is steady or changes very slowly.

- **Range:** minimum to maximum value of input that can be measured. Eg: 0 to 100 deg C.
- **Span:** difference of maximum and minimum of the range. Span = Xmax - Xmin.
- **Accuracy:** closeness of measured value to the true value. Expressed as % of full scale (FS) or % of reading.
- **Precision:** closeness of repeated readings with each other (it is not the same as accuracy).
- **Sensitivity:** change in output per unit change in input = dy/dx. Eg: 10 mV per deg C.
- **Resolution:** the smallest change in input which the sensor can detect.
- **Threshold:** smallest input needed to get any output from zero.
- **Linearity:** how closely the output-input curve is a straight line. Non-linearity = max deviation from best straight line / full scale

output x 100.

- **Hysteresis:** difference in output for the same input when approached from increasing and decreasing directions.
- **Repeatability:** same output for same input under same conditions, in same direction.
- **Reproducibility:** same result when measured by different persons / places / time.
- **Drift:** slow change of output with time at constant input (due to ageing, temperature).
- **Zero error / Offset:** non-zero output at zero input.
- **Dead band (dead zone):** range of input where there is no output change.
- **Stability:** ability to keep its performance over a long time.
- **Error** = Measured value - True value.

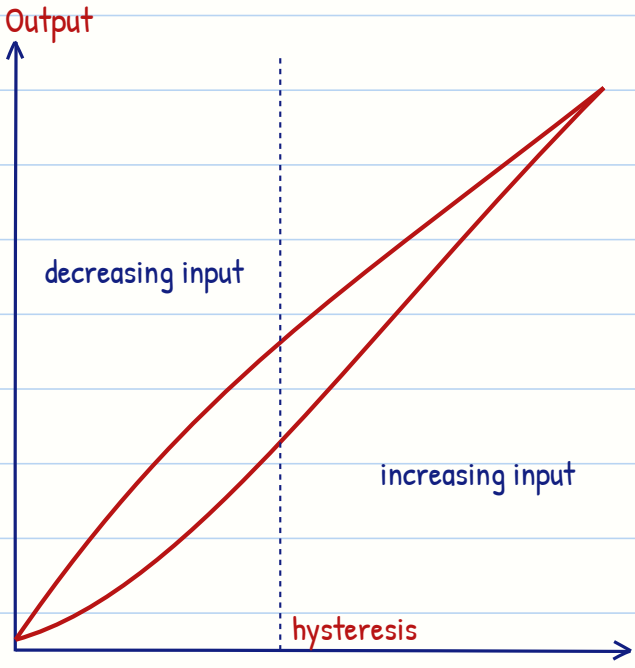


Fig: Hysteresis - different paths for increasing and decreasing input

Accuracy vs Precision (target idea)

- Accurate and precise: all shots at the bullseye. Precise but not accurate: shots grouped but away from centre. Accurate but not precise: shots scattered around the centre.

Calibration

- Comparing the sensor output with a standard of higher accuracy and correcting it. Done periodically to remove zero error and drift.

3. Dynamic Characteristics of Sensors

Dynamic characteristics describe how the sensor responds when the input **changes with time** .

- **Speed of response:** how quickly the output follows the input change.
- **Response time:** time taken to reach a given percentage (say 95% or 98%) of the final value after a step input.
- **Rise time:** time to rise from 10% to 90% of final value.
- **Settling time:** time after which output stays within +/- 2% (or 5%) of final value.
- **Time constant (tau):** time to reach **63.2%** of the final value for a first order system.
- **Lag:** delay of output behind input. **Dynamic error:** difference between true and indicated value when input varies.
- **Fidelity:** ability to reproduce the input variations without distortion.
- **Bandwidth:** range of frequencies over which the sensor responds properly (up to -3 dB).
- **Overshoot, damping ratio and natural frequency** in a second order

sensor (accelerometer, pressure).

Order of the system

- **Zero order:** output follows input instantly. y = K x (potentiometer).
- **First order:** one energy storage, tau dy/dt + y = K x. Step response: y = K x (1 - e^(-t/tau)). Eg: thermocouple, RTD in a thermowell.
- **Second order:** two energy storages, oscillatory. Eg: accelerometer, spring mass damper type sensors, diaphragm pressure sensor.

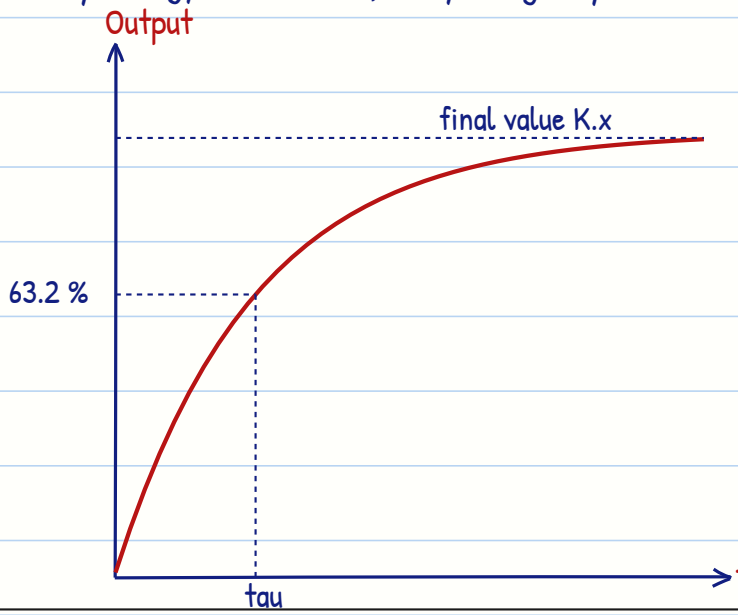


Fig: First order step response - y = K x (1 - e^(-t/tau))

4. Displacement, Position and Proximity Sensors

- **Displacement** = movement of a body from one position to another (linear or angular). **Position** = location of a body w.r.t. a reference. **Proximity** = detecting the presence of an object nearby, without contact.

4.1 Potentiometer (resistive)

- A sliding contact (wiper) moves over a resistive track. Output voltage Vo = Vs x (x / L). Used for linear and rotary displacement.

- Simple and cheap. But friction, wear, noise, limited life and resolution.

4.2 Strain gauge (for small displacement)

- Resistance changes with strain: **Gauge factor GF = (dR/R) / strain** . Used with Wheatstone bridge.

4.3 Capacitive sensor

- Capacitance C = e A / d. Displacement changes the gap d, area A or dielectric. Non contact, high resolution, used for micro displacement and thickness.

4.4 LVDT (Linear Variable Differential Transformer)

- A **primary coil** at the centre and **two secondary coils** wound on either side, connected in series opposition; a movable **ferromagnetic core** inside.
- AC excitation (1 to 10 kHz) in primary. Core at centre: the two secondary voltages are equal and opposite, so Vo = 0 (**null position**).
- Core moves right: V1 > V2, so Vo is proportional to the displacement; **phase** tells the direction (in phase / 180 deg out of phase).
- **Advantages:** frictionless, infinite resolution, high accuracy, robust, range from +/-0.1 mm to +/-250 mm. **Disadvantages:** needs AC supply, demodulator, and is sensitive to stray fields.

Vo = V1 - V2

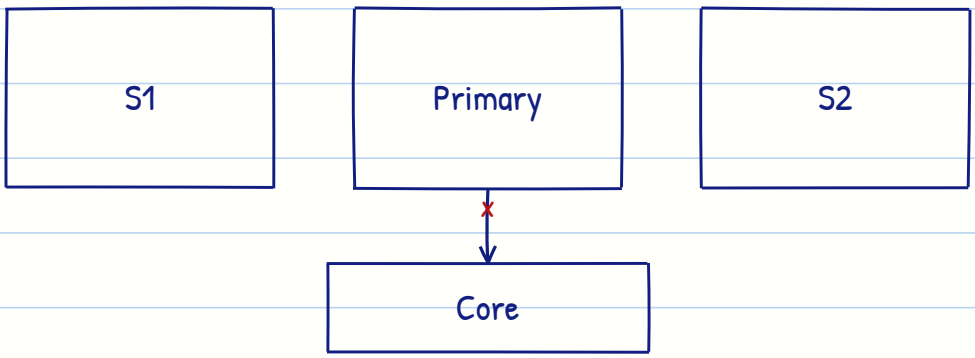


Fig: LVDT - primary, two secondaries and movable core

4.5 Encoders (digital position sensors)

- **Incremental encoder:** a disc with slots, LED and photo detector produce pulses. Counting pulses gives position; two channels A and B (90 deg shifted) give direction; index pulse Z gives reference. Loses position if power fails.
- **Absolute encoder:** disc with coded tracks (binary or **Gray code**) gives a unique code for each position. Position is known directly and is not lost at power off.
- Resolution = 360 / (pulses per revolution). Uses: CNC axes, robot joints, motor speed.

4.6 Resolver and synchro

- Rotating transformers giving sin and cos of shaft angle. Rugged, good for harsh places.

4.7 Proximity sensors

- **Inductive:** detects only **metal** objects; a coil makes an oscillating field,

eddy currents in the metal reduce the oscillation. Range 1 to 30 mm.

- **Capacitive:** detects metals and non metals (plastic, liquid, wood, powder) by change in capacitance.
- **Photoelectric (optical):** LED/laser and photodiode - through beam, retro reflective, diffuse types.
- **Ultrasonic:** transmits sound pulse, measures echo time. Distance = (v x t) / 2.
- **Hall effect:** detects the magnetic field of a magnet. Also **reed switch** (magnetic contact) and limit switches (mechanical).

5. Force and Torque Sensors

5.1 Force sensors

- **Strain gauge load cell:** a metal element (beam, ring, column) deforms under force; gauges bonded on it change resistance. The Wheatstone bridge gives output in mV proportional to the force.
- **Piezoelectric force sensor:** quartz crystal gives charge on loading; good for dynamic force (cutting force) but not static.
- **Capacitive and LVDT type** load cells. **Force sensing resistor (FSR):** resistance falls with force; cheap, used in touch pads and robot grippers.

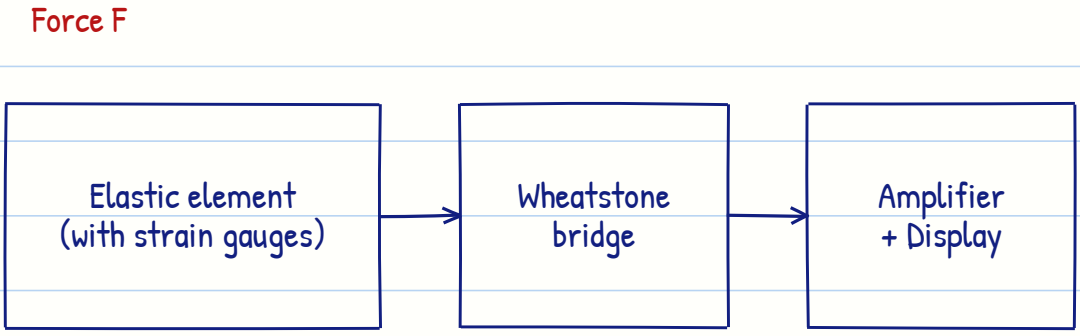


Fig: Strain gauge load cell - block diagram

5.2 Torque sensors

- **Strain gauge torque sensor:** gauges bonded at 45 deg on a shaft measure shear strain due to torsion. Signal taken by slip rings or wireless telemetry.
- **Reaction torque sensor** (static) and **magnetoelastic / optical** non-contact torque sensors.
- Torque T = G J (theta) / L, or from strain: T = (shear strain x G x J) / r.
- Uses: motor testing, robot joints, tightening tools, torque control.

6. Pressure Sensors

- Pressure = Force / Area. Types: absolute, gauge and differential pressure.
- **Elastic types:** diaphragm, **Bourdon tube** , bellows, capsule - the elastic element deflects with pressure. The deflection is converted to an electrical signal by strain gauge, LVDT, capacitive or potentiometer.
- **Strain gauge (piezoresistive) pressure sensor:** gauges on a silicon

diaphragm; widely used, MEMS based.

- **Capacitive pressure sensor:** diaphragm is one plate of a capacitor; pressure changes the gap.
- **Piezoelectric:** quartz - for dynamic pressure, such as in engine cylinders.
- Range: from vacuum to 1000 bar. Used in hydraulic/pneumatic systems, automobile (tyre, MAP), medical (blood pressure).

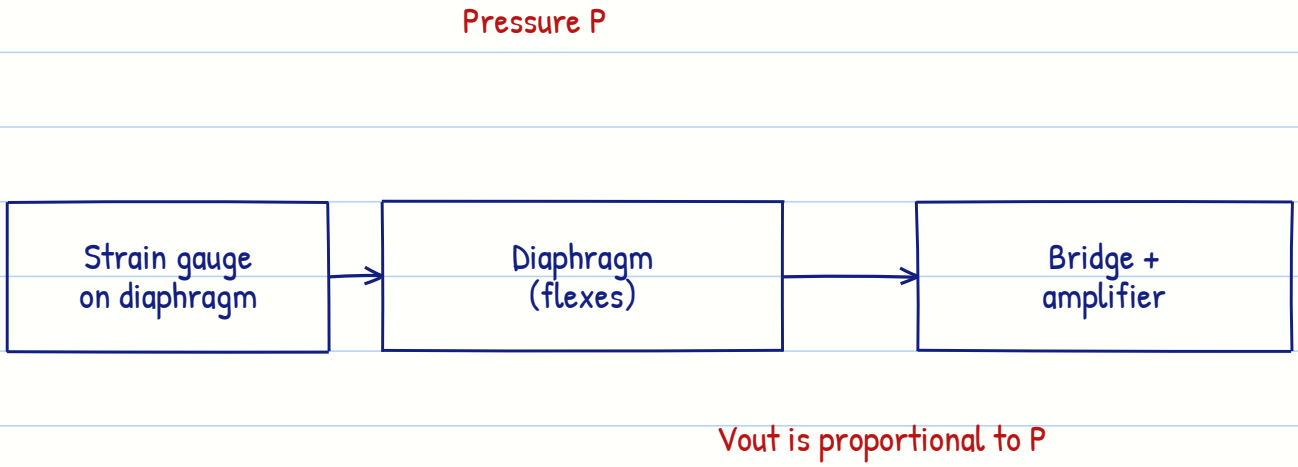


Fig: Diaphragm type pressure sensor

7. Flow Sensors

- Flow rate Q = A x v (m3/s). Used in process plants, fuel systems, water supply and hydraulics.
- **Differential pressure types:** an obstruction like **orifice plate, venturi or flow nozzle** creates a pressure drop. Q is proportional to the square root of the pressure difference (dP). Orifice is cheap, venturi has lower loss.
- **Turbine flow meter:** a rotor spins with speed proportional to flow; a

magnetic pickup gives pulses.

- **Electromagnetic flow meter:** Faraday's law, e = B l v; only for conductive liquids; no obstruction.
- **Ultrasonic flow meter:** transit time or Doppler principle; clamp-on, non invasive.
- **Rotameter** (variable area), **vortex shedding meter** , **coriolis mass flow meter** and **hot wire (thermal) anemometer** for gases.

Type	Principle	Remark
Orifice / Venturi	Pressure drop, Q ~ sqrt(dP)	Cheap, causes pressure loss
Turbine	Rotor speed ~ flow	Good accuracy, clean fluids
Electromagnetic	Faraday's law	Only conducting liquids
Ultrasonic	Transit time / Doppler	Non-invasive, costly

8. Temperature Sensors

- **Thermocouple (Seebeck effect):** two dissimilar metals joined at a hot junction give an emf proportional to the temperature difference with the cold junction. Types: K (Chromel-Alumel, up to 1200 deg C), J (Iron-Constantan), T (Copper-Constantan), R/S (Pt-Rh, up to 1600 deg C). Self generating, wide range, rugged, fast; low voltage so needs cold junction compensation.
- **RTD (Resistance Temperature Detector):** metal wire (platinum **Pt100** , 100 ohm at 0 deg C) whose resistance rises with temperature: R = R0 (1 + alpha T). High accuracy, stable, linear; costly and slower. Range -200 to 650 deg C.

- **Thermistor:** semiconductor (metal oxide); **NTC** type resistance falls strongly with temperature (non linear), **PTC** rises. Very sensitive, cheap, small; limited range (-50 to 150 deg C).
- **IC sensors** (LM35: 10 mV per deg C; AD590) - linear, easy to use. **Bimetallic strip** - mechanical, used in thermostats. **Pyrometer / infrared** - non contact, for very hot or moving objects.

Sensor	Range (deg C)	Output	Features
Thermocouple	-200 to 1700	mV (emf)	Rugged, fast, cheap
RTD (Pt100)	-200 to 650	Resistance change	Accurate, stable
Thermistor	-50 to 150	Resistance change	Very sensitive, non linear
IC (LM35)	-55 to 150	10 mV per deg C	Linear, easy

9. Acceleration Sensors

- Acceleration is measured using a **seismic mass (proof mass)** m on a spring k with damper c inside a case. When the case accelerates, the mass lags and the relative displacement is proportional to acceleration: x = (m a) / k.
- The displacement is sensed by a **piezoelectric crystal** (piezoelectric accelerometer - wide frequency, for vibration), **strain gauge** (piezoresistive) or **capacitive** element (MEMS accelerometer - used in air bags, phones, drones, robots).
- Natural frequency fn = (1/2 pi) sqrt(k/m); the usable range is well below fn.
- Uses: vibration monitoring, crash detection (airbag), navigation, tool

and machine condition monitoring, orientation of mobiles.

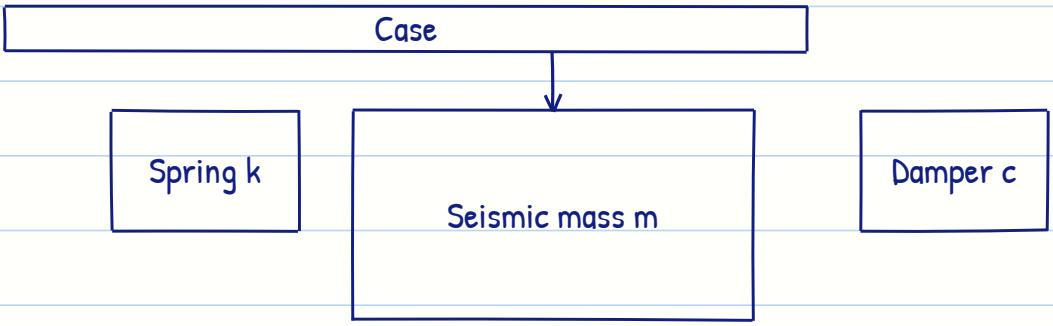


Fig: Seismic mass accelerometer

10. Level Sensors

- Measure the height of liquid or solid in a tank. **Point level** (high / low alarm) and **continuous level** sensors.
- **Float type:** a float moves with the liquid, linked to a potentiometer or a magnetic reed switch.
- **Hydrostatic pressure (differential pressure) type:** P = rho g h, so h = P / (rho g). A pressure sensor at the bottom gives the level.
- **Capacitive level sensor:** probe and tank wall form a capacitor; capacitance rises as liquid covers the probe.
- **Ultrasonic / radar:** echo time from the surface gives distance; non contact.
- **Conductive probes** (for conducting liquids), **optical** (prism) and **load cell** (weight) methods.

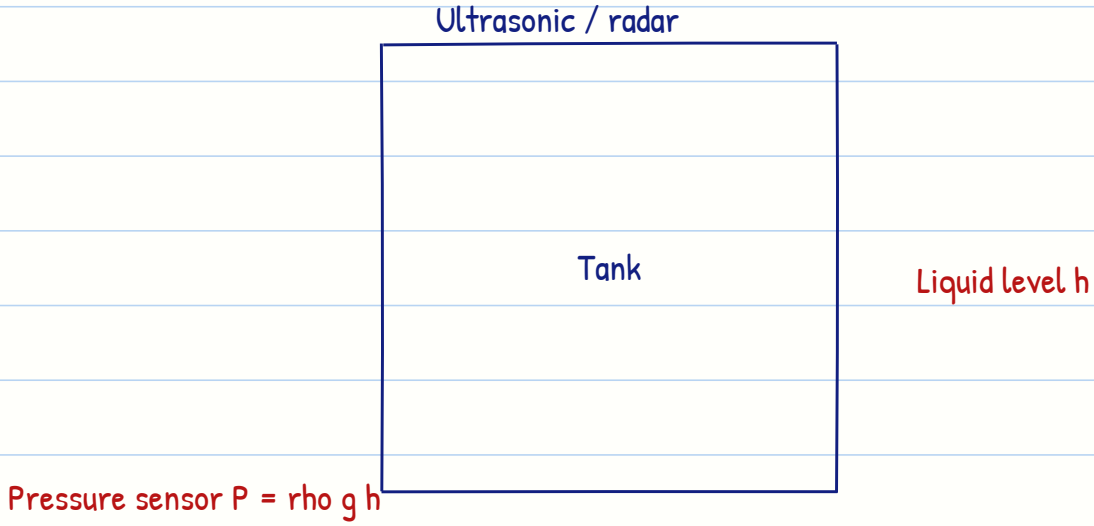


Fig: Level measurement - hydrostatic and ultrasonic methods

11. Light Sensors

- **LDR (Light Dependent Resistor):** CdS material, resistance falls when light intensity rises. Cheap; slow. Used in street lights, alarms.
- **Photodiode:** a reverse biased p-n junction; current is proportional to light. Fast, linear. **Phototransistor:** photodiode with gain.
- **Photovoltaic (solar) cell:** generates voltage from light - self generating.
- **Optical encoders, optocouplers, photo interrupters** and **CCD/CMOS image sensors** (cameras for vision systems).
- **Applications:** object counting, proximity, light curtain safety, vision inspection, bar code readers.

12. Smart Material Sensors

Smart materials change their properties with an external stimulus. They can sense and also actuate.

- **Piezoelectric materials** (quartz, PZT, PVDF): produce charge when stressed (direct effect) and strain when voltage is applied (converse effect). Sensors for force, pressure, vibration, acoustic emission.
- **Magnetostrictive materials** (Terfenol-D): change dimension in a magnetic field; used for position, torque and level sensors.
- **Shape Memory Alloys (Nitinol):** recover shape on heating; temperature sensing.
- **Fibre optic sensors:** light intensity, phase or wavelength changes with strain/temperature (Fibre Bragg Grating). Immune to EMI, good for harsh places, structural health monitoring.
- **Electro and magneto rheological fluids** , **photochromic and thermochromic** materials, **ionic polymer metal composites (IPMC)** and **conducting polymers** .
- **Hall effect and magnetoresistive** (GMR) sensors for magnetic field and speed.

13. Micro and Nano Sensors

- **Micro sensors (MEMS sensors)** are made by silicon micromachining, size 1 micron to 1 mm. Eg: MEMS accelerometer, gyroscope, pressure sensor, microphone, gas sensor, inkjet head.
- **Advantages:** small, light, low power, cheap in mass production, easy integration with electronics (smart sensor), high sensitivity.
- **Nano sensors:** using nanomaterials (**carbon nanotubes, graphene, nanowires, nanoparticles** , quantum dots) of size 1-100 nm. Large surface to volume ratio gives very high sensitivity.

- Examples: gas and chemical nanosensors, biosensors (glucose, DNA, virus), nano pressure and strain sensors, nano mechanical cantilever sensors (detects mass of single molecule).
- **Applications:** automobile (airbag, tyre pressure), medical (implants, blood pressure, glucose), consumer (mobile phone, wearables), industrial IoT, environment monitoring, defence.
- **Challenges:** packaging, noise, calibration, high cost of fabrication setup, and reliability at nano scale.

14. Selection Criteria for Sensors

- **Measurand and range:** the quantity, its minimum and maximum value.
- **Accuracy, precision, resolution** and **sensitivity** needed by the application.
- **Linearity, hysteresis, repeatability** and **drift** .
- **Dynamic response:** response time, bandwidth, natural frequency should suit the signal speed.
- **Environment:** temperature, humidity, dust, vibration, EMI, corrosive fluids, IP rating.
- **Size, weight and mounting;** contact or non-contact; **loading effect** on the measurand.
- **Output type and signal level:** voltage, current (4-20 mA), digital; compatibility with controller and signal conditioning.
- **Power supply and power consumption;** reliability and life; **cost** and availability, ease of maintenance and calibration.

Important Questions (Unit II)

- Explain static and dynamic characteristics of sensors.
- Explain the working of LVDT / encoders with neat sketches.
- Explain proximity sensors and their types.
- Explain temperature sensors - thermocouple, RTD and thermistor.
- Write short notes on force, pressure, flow, level, light and acceleration sensors.
- Write about smart material sensors, micro and nano sensors and the selection criteria for sensors.

MECHATRONICS AND MEMS

Subject Code : 23A03702d

UNIT III : Actuators

Syllabus: Mechanical, Electrical, Hydraulic and Pneumatic Actuation systems, Characteristics and their limitations, Design of Hydraulic and Pneumatic circuits, Piezoelectric actuators, Shape memory alloys, Selection criteria for actuators.

I. Introduction to Actuators

- An **actuator** is the output device of a mechatronic system. It takes the low power control signal from the controller and converts it into **physical action** - linear or rotary motion, force, torque, heat - to drive the load.
- Actuator = **power converter** (electrical / fluid energy -> mechanical energy). The controller output is amplified by a **driver** (relay, transistor, SCR, servo amplifier, valve).
- **Types of actuation systems**
  - **Mechanical** (gears, cams, belts, linkages)
  - **Electrical** (solenoids, relays, DC / AC / stepper / servo motors)
  - **Hydraulic** (oil under high pressure)
  - **Pneumatic** (compressed air)
  - **Smart material actuators** (piezoelectric, shape memory alloy, magnetostrictive)

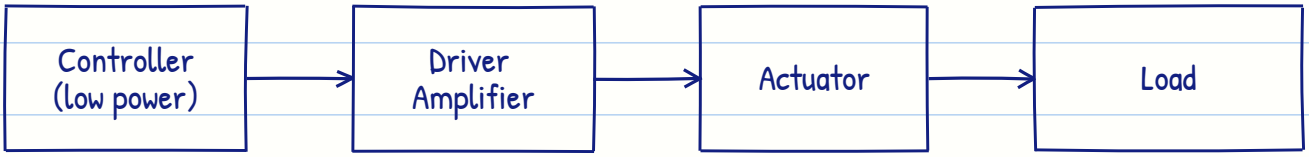


Fig: Actuator in a mechatronic system

2. Mechanical Actuation Systems

Mechanical systems transmit and change motion, speed, torque and direction between the prime mover and the load.

2.1 Gears and gear trains

- Change speed and torque. Gear ratio G = N1 / N2 = T2 / T1 = z2 / z1. A speed reducer increases torque.
- Types: spur, helical, bevel, worm (high reduction, self locking), rack and pinion (rotary to linear), planetary and harmonic drives (robots, compact, no backlash).
- **Backlash** is the play between meshing teeth - it causes positioning error; it is reduced by anti-backlash gears.

2.2 Belt and chain drives

- Flat, V, **timing (toothed) belts** and chains transmit power over long distance. Timing belt has no slip, so used in 3D printers and CNC.

2.3 Cams and followers

- A cam converts rotary motion into a precise, programmed reciprocating motion of the follower. Types: disc, cylindrical, translating cams. Used in engines (valve operation), packaging and automatic machines.

2.4 Ratchet and pawl, Geneva mechanism, linkages

- **Ratchet and pawl:** one directional motion; **Geneva wheel:** gives intermittent (indexing) rotation; **four bar linkage and slider crank:** convert rotation to oscillation or reciprocation.

2.5 Bearings, lead and ball screws

- **Lead screw / Ball screw and nut:** convert rotary motion to linear motion. Ball screw has recirculating balls so friction is very low and efficiency is 90%+; used in all CNC axes. Linear displacement = lead x revolutions.
- **Bearings** (ball, roller, linear guides) support the load and reduce friction.

Characteristics and limitations

- Reliable, simple and robust, give high torque. But they have **backlash, friction, wear** , need lubrication, are bulky, and the motion is fixed (not flexible) except for cams that need changing.

3. Electrical Actuation Systems

Electrical actuators are the most common - they are clean, easy to control, accurate, and compatible with electronics.

3.1 Switching devices

- **Mechanical switches and relays:** electromagnetic relay - a small current in the coil closes contacts of a heavy load circuit. Isolation, but slow and contacts wear.
- **Solid state relay (SSR), diode, thyristor (SCR), triac, BJT, MOSFET, IGBT:** electronic switches for fast, silent, long life switching and speed control of motors.

3.2 Solenoids

- A coil with a movable iron plunger. When current flows, the magnetic field pulls the plunger in (linear motion). Spring returns it back. Force is proportional to (N I)^2.
- Used in solenoid valves, door locks, relays, printers. Short stroke, on/off action only.

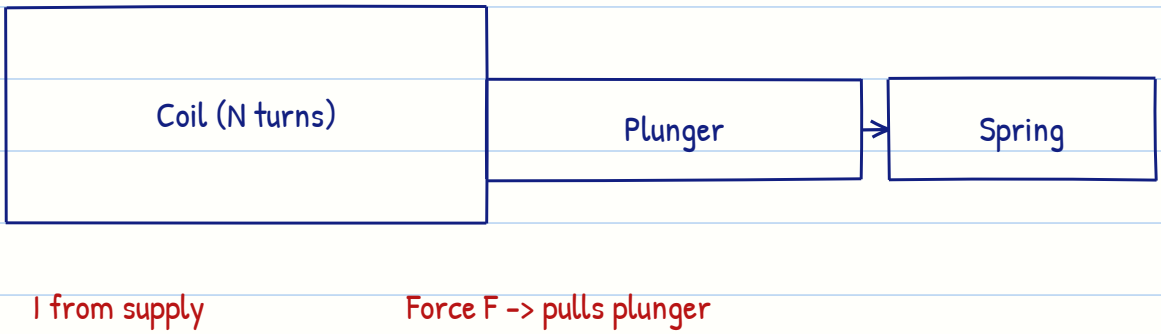


Fig: Solenoid actuator

3.3 DC motors

- Torque T = K phi Ia. Back emf Eb = K phi w. Speed w = (V - Ia Ra) / (K phi).
- **Permanent magnet DC (PMDC), separately excited, series and shunt** types. Speed control by armature voltage using **PWM** with H-bridge (also gives reversing).
- Advantages: good starting torque, easy speed control, linear torque-speed characteristic. **Limitations:** brushes and commutator wear, sparking, noise, maintenance.
- **Brushless DC motor (BLDC):** permanent magnet rotor, electronic commutation using hall sensors; no brush wear, high efficiency, long life, higher cost. Used in drones, EV, hard disks.

3.4 AC motors

- **Induction motor:** rugged, cheap, little maintenance; speed Ns = 120 f / P. Speed control using **VFD (variable frequency drive)** V/f control. Used in conveyors, pumps, machine spindles.
- **Synchronous motor:** runs at exact synchronous speed; used with servo drives (PMSM).

3.5 Stepper motors

- Rotate in fixed **steps** per pulse (1.8 deg per step = 200 steps/rev). Open loop position control; no feedback needed.
- Types: **permanent magnet, variable reluctance and hybrid** . Step angle = 360 / (number of steps). Speed (rpm) = (step angle x pulse rate x 60) / 360.
- Driven by a driver using full step, half step or micro stepping.

  Advantages: precise, cheap, holding torque, simple control. **Limitations:** loss of steps at high load or speed, vibration/resonance, low efficiency, torque falls with speed.
- Uses: 3D printers, CNC small machines, camera, printers, robots.

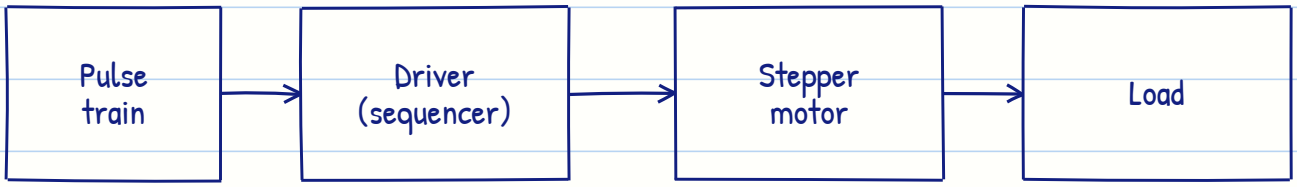


Fig: Stepper motor control (open loop)

3.6 Servo motors

- A **servo motor** is a motor + position/speed sensor (encoder or resolver) + controller in **closed loop** . It follows the command position exactly, with fast response and high torque at high speed.
- Types: DC servo and AC servo (PMSM / BLDC). Control: position loop, velocity loop and current loop (cascade PID).
- **Advantages:** high accuracy, high speed, good torque to inertia ratio, handles load changes. **Limitations:** costlier than stepper, needs tuning, feedback device needed, complex.

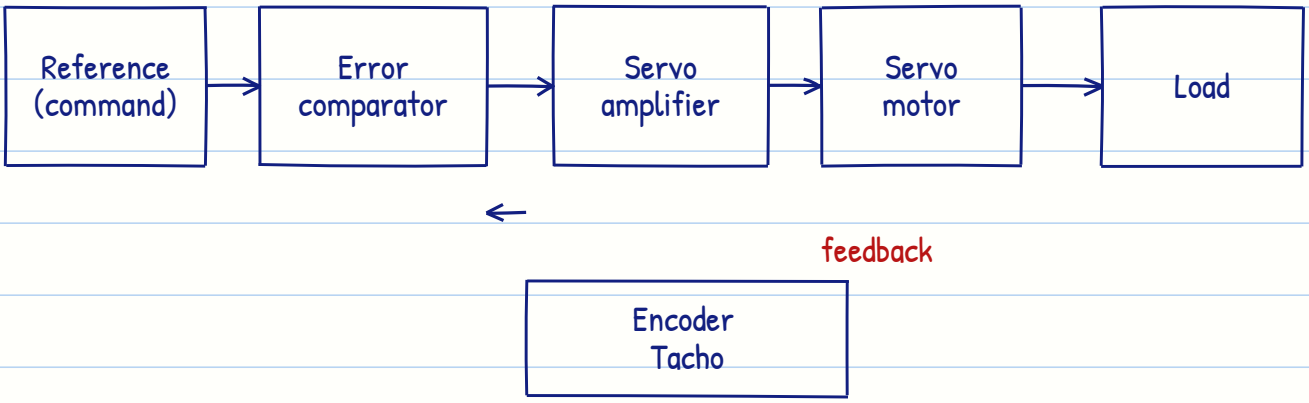


Fig: Closed loop servo system

Point	Stepper motor	Servo motor
Control	Open loop (no feedback)	Closed loop (encoder)
Cost	Low	High
Speed / torque	Torque falls fast with speed	Good torque at high speed
Accuracy	Good, may lose steps	Very high, no step loss
Use	3D printer, small CNC	CNC, robots, high performance

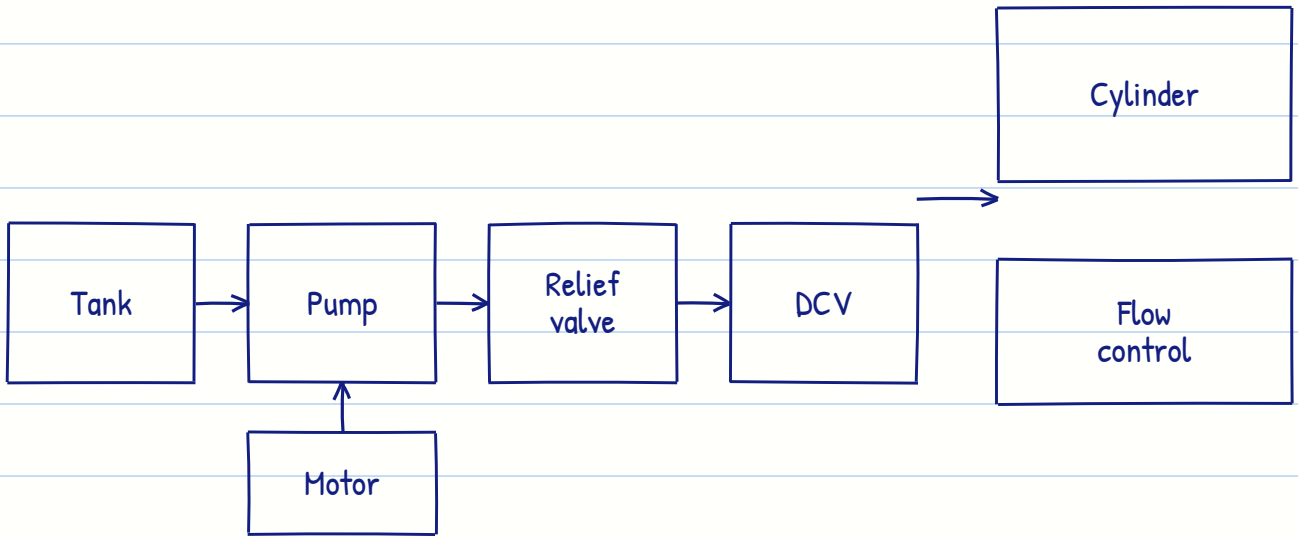
4. Hydraulic Actuation Systems

- Use **incompressible liquid (mineral oil)** at high pressure (up to 350 bar) to transmit power. Based on **Pascal's law:** pressure applied to a confined fluid is transmitted equally in all directions. $F = P \times A$.
- Gives **very large forces** from small size, smooth and stiff motion, good for heavy loads - presses, excavators, aircraft controls, machine tools, robots (large).

Components of a hydraulic system

- **Reservoir (tank):** stores oil, cools and settles dirt.
- **Pump:** converts mechanical energy to fluid energy. Gear, vane and **piston pumps** (positive displacement), driven by electric motor.
- **Pressure relief valve:** safety valve limits maximum pressure.
- **Filters and strainers,** coolers, accumulators and pressure gauges.
- **Control valves:** direction control valve (DCV), pressure control and flow control (needle) valves.
- **Actuators: cylinders** (linear: single acting, double acting) and **hydraulic motors** (rotary).
- Pipes, hoses and fittings.

Fig - Basic hydraulic power pack



Hydraulic cylinders

- **Single acting:** pressure on one side only, spring or load returns it. **Double acting:** pressure on both sides - force both ways. Force extend $F = P \times A$; retract $F = P \times (A - a)$ (rod area a). Speed $v = Q / A$.

Hydraulic directional control valves

- Described by **ways and positions** - eg: **4/3 valve** (4 ports, 3 positions) with centre positions closed, open or tandem. Operated by lever, solenoid, pilot pressure; **servo valves** and **proportional valves** give continuous control.

Advantages

- Large force and power to weight ratio, easy speed control, smooth, self lubricating, can be stopped under load, overload protection by relief valve.

Limitations

- Oil **leakage and mess** , fire hazard, needs pump, cooling and filtering, expensive, noisy, viscosity changes with temperature, contamination and

maintenance, slower than electric / pneumatic.

5. Pneumatic Actuation Systems

- Use **compressed air** (4 to 10 bar, normally 6 bar) to drive actuators. Air is available free, clean, safe (no fire or spark) and the exhaust goes to atmosphere (no return line).
- Used for fast on/off motions - clamping, pick and place, packaging, door opening, tools (drill, hammer), automation.

Components

- **Compressor:** piston/screw/vane compressors, **air receiver (tank)** , after cooler and dryer.
- **FRL unit (Filter, Regulator, Lubricator)** - service unit that cleans, controls pressure and oils the air.
- **Control valves:** DCV (3/2, 5/2, 5/3), **flow control** (speed control by throttling), **non return, quick exhaust, shuttle (OR) and two pressure (AND)** valves.
- **Actuators: pneumatic cylinders** (single acting, double acting, rodless, rotary), air motors.

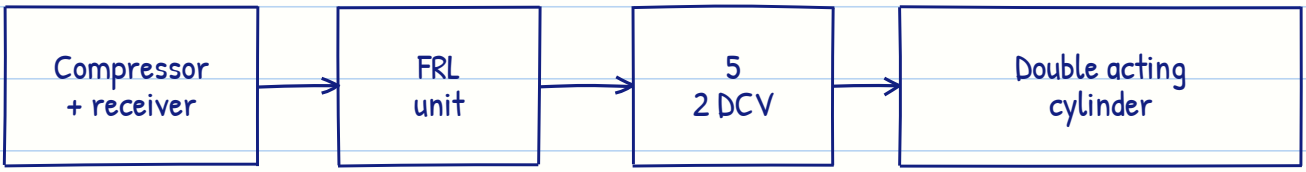


Fig: Basic pneumatic system

Pneumatic vs Hydraulic

Point	Hydraulic	Pneumatic
Medium	Oil	Compressed air
Pressure	Up to 350 bar	4 to 10 bar
Force	Very high	Low to medium
Speed	Slow, smooth	Fast, jerky
Compressibility	Incompressible, stiff	Compressible, spongy
Leakage	Messy oil	Harmless air
Cost	High	Low
Return line	Required	Not needed (exhaust)

Limitations of pneumatic

- Compressible air gives poor position control and low stiffness; limited force; noise of exhaust; moisture in air; energy loss in air generation.

6. Design of Hydraulic and Pneumatic Circuits

- Circuits are drawn using **ISO 1219 graphical symbols** . Draw supply at

the bottom, control at the middle, actuator at top, cylinder on the right. Label components (1.0 cylinder, 1.1 valve, 0.1 supply).

Steps of circuit design

- Understand the task (load, stroke, speed, sequence); select cylinder size from F = P A (add 10-25% for friction); choose valve type and control method (manual, solenoid, pilot); add speed control and safety (relief, check valve); select pump / compressor, tubing, filter; draw the circuit and test.

6.1 Control of a single acting cylinder

- A **3/2 DCV** is used. Pressed: air goes to cylinder, it extends. Released: spring returns the valve, air exhausts, the cylinder spring returns the piston.

6.2 Control of a double acting cylinder

- A **4/2 or 5/2 DCV** (hydraulic: 4/3 valve) - piston extends with one position and retracts with the other. Speed control by **flow control valves** in meter-in or meter-out. Meter-out is preferred for stable speed.

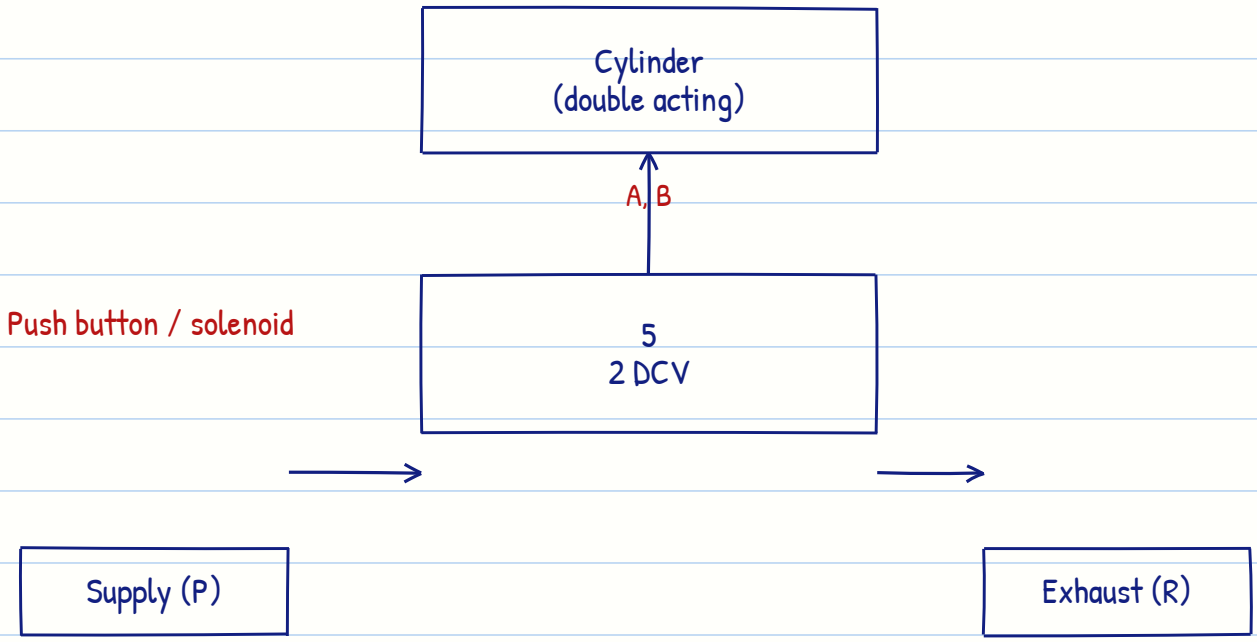


Fig: Double acting cylinder control with 5/2 valve

6.3 Speed control and special circuits

- **Meter-in:** throttle in supply line; **meter-out:** throttle in exhaust line; **bleed-off:** part of oil goes back to tank (hydraulic).
- **Regenerative circuit:** oil from the rod side is joined with the pump flow - fast extension. **Sequencing / counter balance valve** keeps a vertical load from falling. **Pressure reducing valve** for a lower clamp pressure.
- **AND (two pressure) valve:** safety with two hands. **OR (shuttle) valve:** operate from either of two places.

6.4 Sequence circuit (A+ B+ B- A-)

- Two cylinders A and B: A extends, then B extends, B retracts, and A retracts. **Limit switches / roller valves** at the ends give the signal for the next step; use **cascade or electro pneumatic (relay / PLC)** control to avoid overlapping signals.

6.5 Hydraulic circuit example

- A press circuit: pump -> relief valve -> 4/3 DCV (tandem centre, unloads

pump) -> double acting cylinder, with a counterbalance valve and a pressure gauge.

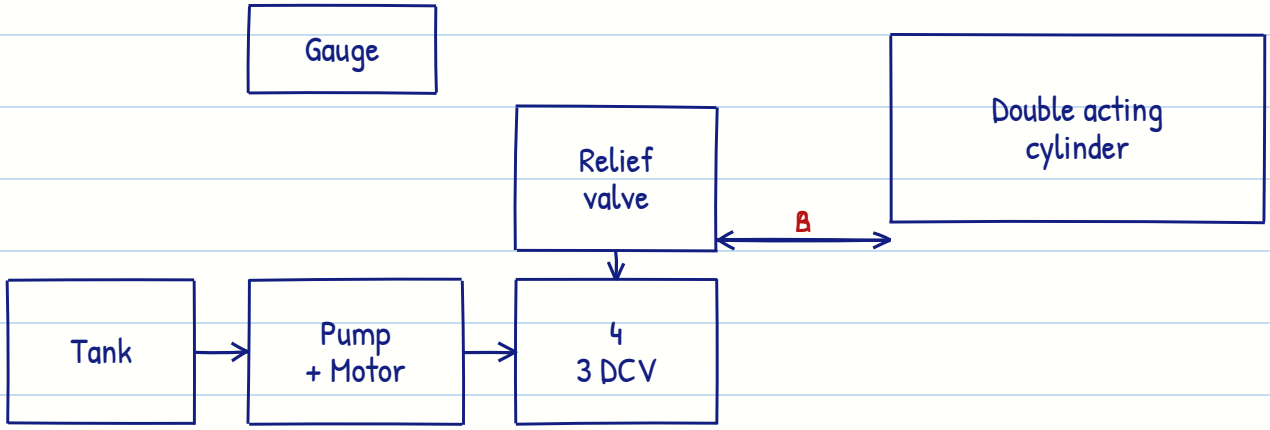


Fig: Hydraulic circuit with 4/3 DCV

7. Piezoelectric Actuators

- **Piezoelectric effect:** certain crystals (quartz, **PZT – lead zirconate titanate** , PVDF) develop an electric charge on mechanical stress (direct effect – sensor). The **converse effect:** they change shape when voltage is applied – used as actuator.
- A stack of thin PZT layers (multilayer stack) gives larger displacement. Strain is about 0.1% – so displacement is only a few microns to tens of microns; **bender and amplified** types give more stroke.
- Displacement dL = d33 x V (for stack with n layers, x n). Response time in microseconds, force up to kN.

Characteristics

- **Very high resolution (nm), fast response (kHz), high force, no**

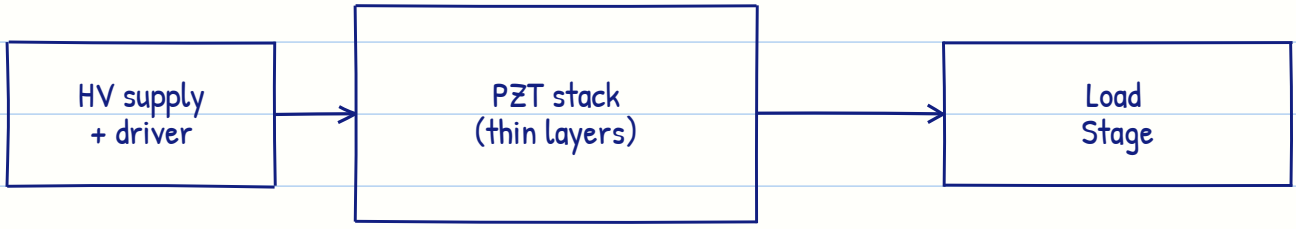
magnetic field, no backlash, and low power at hold. Works in vacuum / clean room; high stiffness.

Limitations

- **Small stroke,** needs **high voltage** (100-1000 V) drive, hysteresis and creep, brittle ceramic, cannot take tension, cost, temperature limit (Curie point).

Applications

- Inkjet printer heads, fuel injectors (diesel/gasoline), nano positioning stages, atomic force microscope (AFM), ultrasonic motors and cleaners, autofocus, vibration damping, micro pumps, MEMS.



V up -> length increases (micro displacement)

Fig: Piezoelectric stack actuator

8. Shape Memory Alloys (SMA)

- SMA are metals which **remember their original shape** . After deformation at low temperature they recover the shape on **heating** above the transformation temperature. Eg: **Nitinol (Nickel-Titanium, 55% Ni - 45% Ti)** , Cu-Zn-Al, Cu-Al-Ni.
- **Principle:** solid state phase change between **martensite** (low temperature, soft, easily deformed) and **austenite** (high temperature,

stiff, original shape). Transformation temperatures: Ms, Mf (martensite start / finish) and As, Af (austenite start / finish).

- **One way memory effect:** deform in martensite -> heat -> recovers the shape; on cooling it does not change shape back until deformed again. **Two way:** remembers both hot and cold shapes. **Superelasticity (pseudo elasticity):** recovers large strain (8%) on unloading at constant temperature.
- **Heating** is done by passing an electric current through the wire (Joule / resistance heating) - so it acts as an electrical actuator. Cooling by air.

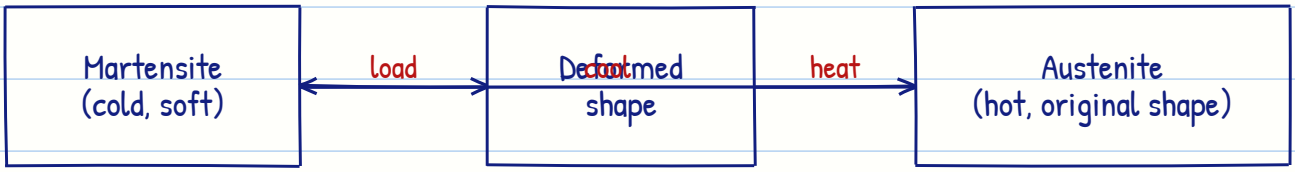


Fig: Shape memory effect cycle

Characteristics

- Recovery strain up to 4-8%, **high force per unit weight** , silent, compact, simple, biocompatible, corrosion resistant.

Limitations

- **Slow** (cooling is slow, 1-few Hz), low efficiency (about 5%), hysteresis, fatigue (limited life), complex control, expensive, small strokes.

Applications

- Robot grippers and artificial muscles, micro actuators, valves, thermostat,

couplings, aerospace (morphing wing), medical stents and guidewires, dental braces, eyeglass frames, fire safety.

9. Other Smart Actuators (brief)

- **Magnetostrictive actuators** (Terfenol-D): change length in a magnetic field; high force, fast; used in sonar and precision positioning.
- **Electro / Magneto-rheological fluids:** viscosity changes under field - dampers, clutches, brakes.
- **Electro active polymers (EAP), IPMC:** artificial muscles; **electrostrictive** and **thermal bimorph** actuators in MEMS.
- **Electrostatic (comb drive) actuators** in MEMS - covered in Unit V.

10. Selection Criteria for Actuators

- **Load requirement:** force or torque, speed, stroke / angle, power, duty cycle and load inertia (match inertia ratio for motors).
- **Accuracy, resolution and repeatability** - stepper for light open loop, servo for high accuracy, piezo for nanometre.
- **Speed and response (bandwidth):** pneumatic fast on/off, hydraulic smooth, servo fast.
- **Power source available:** electrical supply, compressed air or hydraulic power pack.
- **Controllability and compatibility** with the controller (PWM, pulse, voltage, current signal).
- **Size, weight, and mounting;** environment (explosive, dust, temperature, wash down).
- **Efficiency, power consumption, noise, vibration;** safety.

- **Reliability, life and maintenance; cost** (initial + running) and availability.

Actuator	Force / torque	Accuracy	Typical use
Pneumatic	Low-medium	Low (on/off)	Pick & place, clamping
Hydraulic	Very high	Medium	Press, excavator
Stepper	Low-medium	Good (open loop)	3D printer, small CNC
Servo	Medium-high	Very high	CNC, robots
Piezo	High but tiny stroke	nm level	Nano stage, injector
SMA	High force / weight	Low, slow	Micro gripper, valve

Important Questions (Unit III)

- Explain mechanical, electrical, hydraulic and pneumatic actuation systems with their limitations.
- Compare stepper and servo motors. Explain the working of a stepper motor.
- Compare hydraulic and pneumatic systems. Explain the components of a hydraulic system.
- Design a pneumatic circuit for a double acting cylinder / sequence A+ B+ B- A-.
- Explain piezoelectric actuators and shape memory alloys with applications.
- What are the selection criteria for actuators?

MECHATRONICS AND MEMS

Subject Code : 23A03702d

UNIT IV : Microprocessors, Microcontrollers and PLCs

Syllabus: Architecture of Microprocessor, Microcontroller and Programmable Logic Controller, PLC Programming using ladder diagrams, logics, latching, sequencing, timers relays and counters, data handling, Analog input/output, selection of controllers.

1. Controllers in Mechatronics

- The **controller** is the brain of a mechatronic system. It reads the inputs (sensors), runs a program / logic, and gives outputs to actuators.
- Common controllers: **Microprocessor (uP) based systems, Microcontroller (uC), Programmable Logic Controller (PLC)** , also DSP, FPGA and industrial PC.
- Number systems used: binary, hexadecimal. Data bus width = 8, 16, 32 bits.

2. Architecture of a Microprocessor

- A **microprocessor** is a single chip CPU (VLSI). It needs **external memory and I/O chips** to make a complete system - so it is general purpose and flexible. Eg: Intel 8085 (8 bit), 8086 (16 bit), Pentium.

Main blocks of the microprocessor

- **ALU (Arithmetic Logic Unit):** does arithmetic (add, subtract) and

logic (AND, OR, XOR, compare) operations.

- **Registers:** fast internal storage - **accumulator** (result of ALU), general purpose registers (B, C, D, E, H, L), **program counter PC** (address of next instruction), **stack pointer SP** , instruction register, flag register (zero, carry, sign, parity).
- **Control unit:** fetches, decodes and executes instructions; generates timing and control signals.
- **Buses:**
  - **Address bus** (unidirectional) - carries the memory/IO address. 16 lines can access 2^16 = 64 KB.
  - **Data bus** (bidirectional) - carries the data (8 bit).
  - **Control bus** - signals such as RD, WR, ALE, IO/M, interrupt (INTR, TRAP).

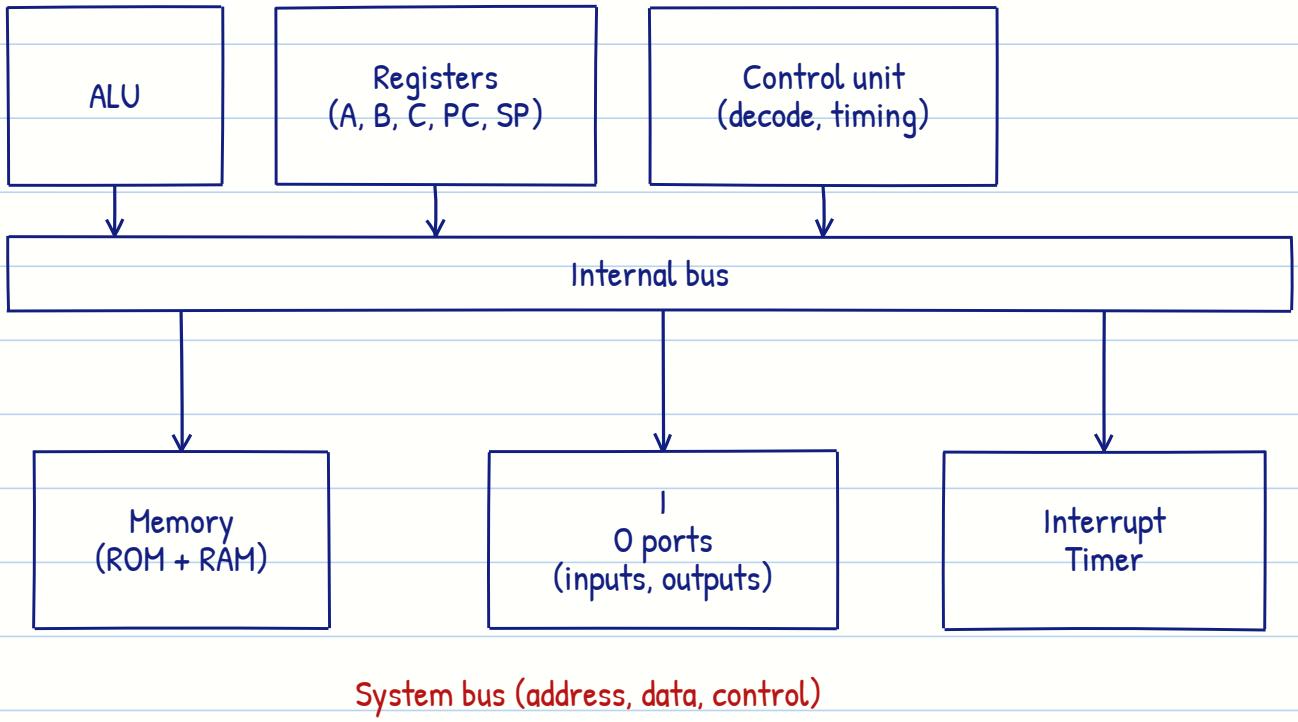


Fig: Microprocessor based system - CPU with external memory and I/O

Instruction cycle (fetch – decode – execute)

- **Fetch:** PC puts the address on the address bus, instruction is read from memory into the instruction register, PC = PC + 1. **Decode:** control unit decodes the opcode. **Execute:** ALU / registers / memory perform the operation. Then the next fetch.
- 8085 features: 8 bit data, 16 bit address, 5 V single supply, clock up to 3 MHz, 74 instructions, 5 interrupts, 40 pin.

Von Neumann vs Harvard architecture

- **Von Neumann:** single memory and bus for program and data (microprocessors, PC). **Harvard:** separate program and data memories and buses – faster, used in microcontrollers (PIC, AVR, 8051) and DSP.

RISC and CISC

- **CISC:** many complex instructions (x86). **RISC:** small set of simple instructions, one cycle each, pipelining (ARM, AVR, PIC).

3. Architecture of a Microcontroller

- A **microcontroller** is a **computer on a single chip** : CPU + RAM + ROM/Flash + I/O ports + timers/counters + serial port + interrupt + ADC etc. all in one IC. It is **dedicated to one control task** (embedded system). Eg: Intel 8051, PIC16F877, AVR ATmega328 (Arduino), ARM Cortex, ESP32.

Block diagram (8051 as example)

- **CPU (8 bit) with ALU, accumulator A, B register, PSW, PC, DPTR, SP.**
- **ROM / Flash:** 4 KB on-chip program memory. **RAM:** 128 bytes data

memory (register banks, bit addressable area, scratch pad).

- **I/O ports:** 4 ports P0, P1, P2, P3 (32 I/O lines). P0 and P2 also work as multiplexed address/data bus when external memory is used.
- **Timers/counters:** two 16 bit (T0, T1) - for delay, counting events, baud rate. **Serial port:** UART full duplex. **Interrupts:** 5 sources (2 external, 2 timers, 1 serial). Oscillator / clock circuit (11.0592 MHz).

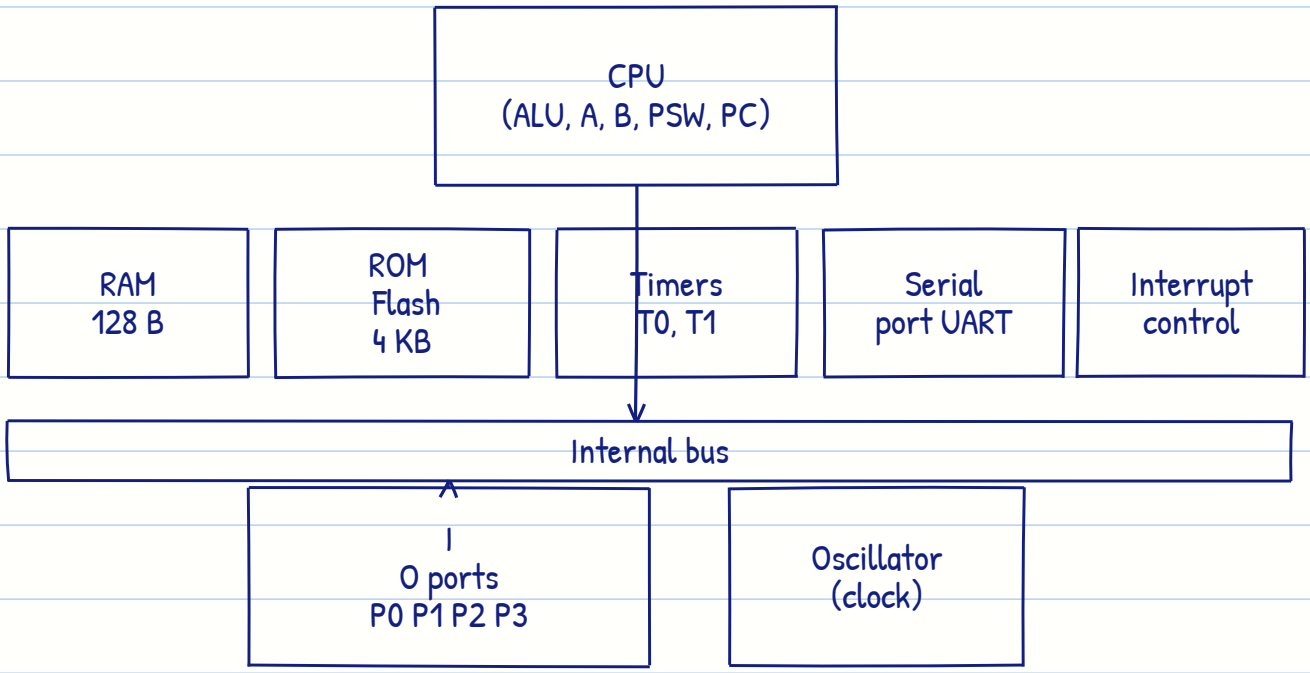


Fig: Block diagram of 8051 microcontroller

Microprocessor vs Microcontroller

Point	Microprocessor	Microcontroller
Chip contents	Only CPU	CPU + RAM + ROM + I/O + timer
External parts	Needs memory, I/O chips	Very few (all inside)
Cost and size	Higher, bigger board	Low, compact
Power	High	Low

Speed	High (GHz)	Lower (MHz)
Use	General purpose (computer)	Dedicated (washing machine, car ECU)
Architecture	Von Neumann mostly	Harvard mostly
Bit manipulation	Few instructions	Many bit-level instructions

Applications of microcontrollers in mechatronics

- Motor speed / position control, CNC and robot joints, automobile ECU (engine, ABS), washing machine, microwave oven, remote controls, smart meters, 3D printer, drones, IoT nodes.

4. Programmable Logic Controller (PLC)

- **Definition (NEMA):** A PLC is a digital electronic device with **programmable memory** to store instructions that implement functions like logic, sequencing, timing, counting and arithmetic to control machines and processes through digital or analog inputs and outputs.
- Designed for the **harsh industrial environment** - dust, noise, vibration, EMI, temperature. First used in 1968 by General Motors to replace relay panels (Modicon 084, Dick Morley).
- Eg: Siemens S7, Allen-Bradley (Rockwell), Mitsubishi FX/Q, Omron, Delta, ABB.

Advantages over relay logic

- Changes by software, no rewiring; compact; reliable (no contacts to wear); fast; easy fault finding (indicator LEDs, diagnostics); counting, timing, arithmetic and communication; low cost per function; program can be stored and copied.

Architecture of a PLC

- **Central Processing Unit (CPU):** microprocessor that executes the user program in a repeated **scan cycle** ; has the system program (operating system).
- **Memory: ROM** (system program), **RAM** (user program, data, I/O image table; battery backup), **EEPROM / Flash** (permanent program storage).
- **Input module:** takes field signals (push buttons, limit switches, sensors) 24 V DC / 230 V AC, with **opto isolation** , filter and level shifting to logic levels.
- **Output module:** gives power to loads (contactor, solenoid valve, lamps, motor starters). Types: **relay** (AC/DC, slow, high current), **transistor** (DC, fast), **triac** (AC, fast).
- **Power supply:** converts 230 V AC to 24 V DC / 5 V for the CPU and modules.
- **Programming device:** PC with software / handheld programmer. **HMI/SCADA** and communication ports (RS232, RS485, Ethernet, Profibus, Modbus).

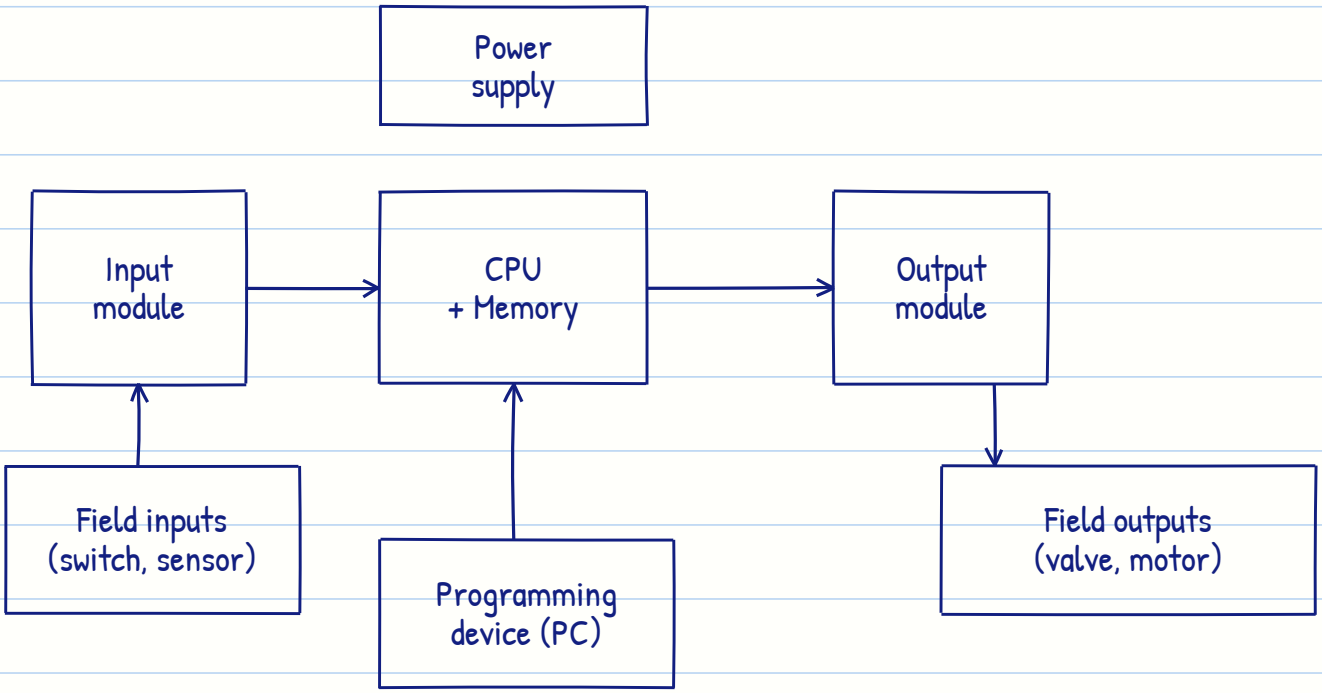


Fig: Architecture of a PLC

Scan cycle of a PLC

- **1. Read inputs** (copy the input states to the input image table) -> **2. Execute the program** (rung by rung, left to right, top to bottom) -> **3. Update outputs** (write the output image table to output terminals) -> **4. Housekeeping / communication, self diagnosis** . Then repeat. Scan time is typically 1 to 100 ms.

Types of PLC by size

- **Compact / fixed (nano, micro):** all in one box, up to 128 I/O - small machines. **Modular:** rack, CPU and plug in modules - **large** , more than 1000s of I/O, expandable.

Input / Output devices

- Inputs: push buttons, selector switches, limit switches, proximity and photo sensors, pressure switches. Outputs: relay, contactors, solenoid valves, indicator lamps, buzzers, motor drives.

5. PLC Programming Languages (IEC 61131-3)

- **Ladder Diagram (LD)** - graphical, like relay circuit; most popular. **Function Block Diagram (FBD)** - blocks. **Structured Text (ST)** - high level like Pascal. **Instruction List (IL)** - assembly like mnemonics (LD, AND, OR, OUT). **Sequential Function Chart (SFC)** - steps and transitions.

6. PLC Programming using Ladder Diagrams

- A ladder diagram has two vertical **power rails** (left = live, right = neutral/return) and horizontal **rungs** between them, looking like a ladder. Power "flows" left to right when the logic of a rung is TRUE (**power flow**). Each rung has **inputs (conditions) on the left** and **one output (coil) on the right.**

Symbols

- **Normally Open contact (NO) -| |-:** True when the input is ON (1). **Normally Closed contact (NC) -|/|-:** True when the input is OFF (0). **Output coil -( )-** . **Set / Reset coil** (latch/unlatch). Addresses: I0.0, X0 (input), Q0.0, Y0 (output), M (memory), T (timer), C (counter).

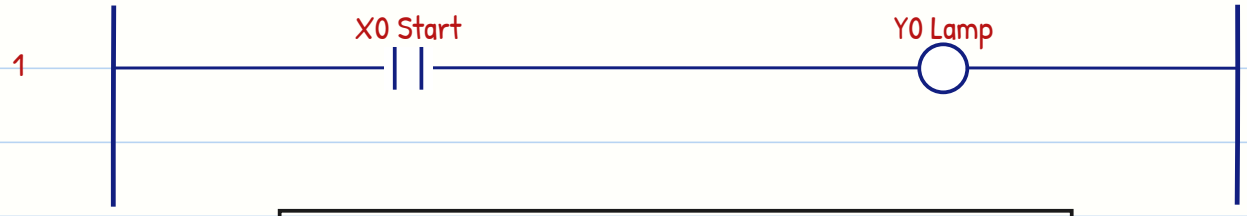


Fig: Rung 1 - simple ON action: Y0 = X0

6.1 Logic functions in ladder

- **AND:** contacts in **series** - output ON only when all the contacts are closed. **OR:** contacts in **parallel** - output ON if any contact closed.

NOT: use NC contact. **NAND, NOR, XOR** are obtained by combining these.

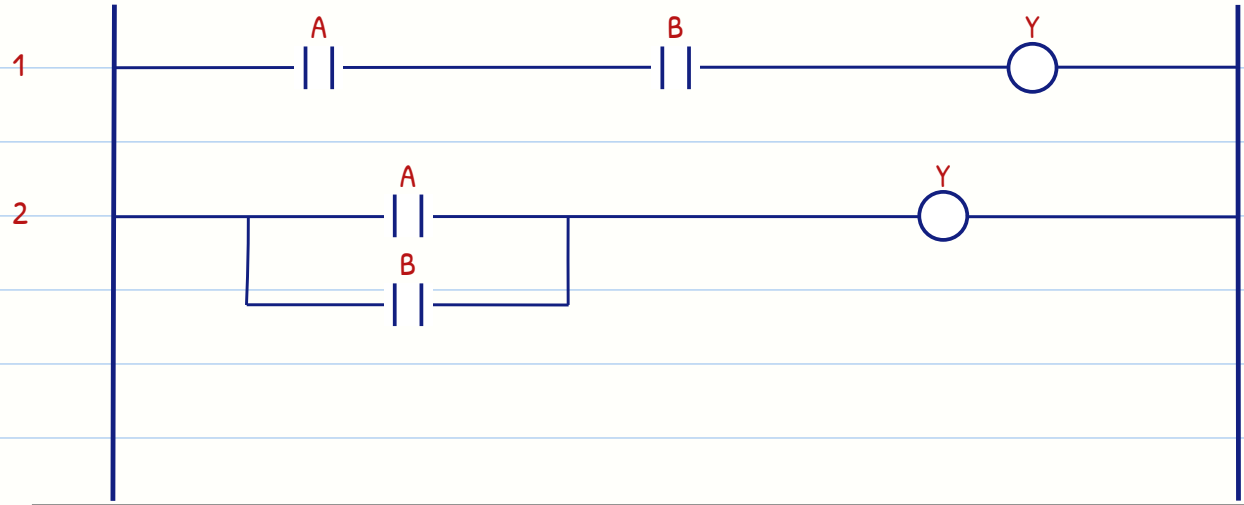


Fig: AND (series) and OR (parallel) logic - rung 1 AND, rung 2 OR

- **XOR:** Y = A B' + A' B -> two parallel branches: (A NO in series with B NC) OR (A NC in series with B NO).
- **NOT:** NC contact. **NAND:** parallel of NC contacts. **NOR:** series of NC contacts.

A	B	AND	OR	XOR
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

6.2 Latching (self-holding / sealing circuit)

- A push button gives only a momentary signal. To keep the output ON after the button is released, a **holding (sealing) contact** of the output itself is placed in parallel with the start button. **Stop button (NC)** is in series to break the circuit.
- Y0 = (Start + Y0) . Stop'. Used for motor start/stop control (DOL

starter), lamp control, alarms.

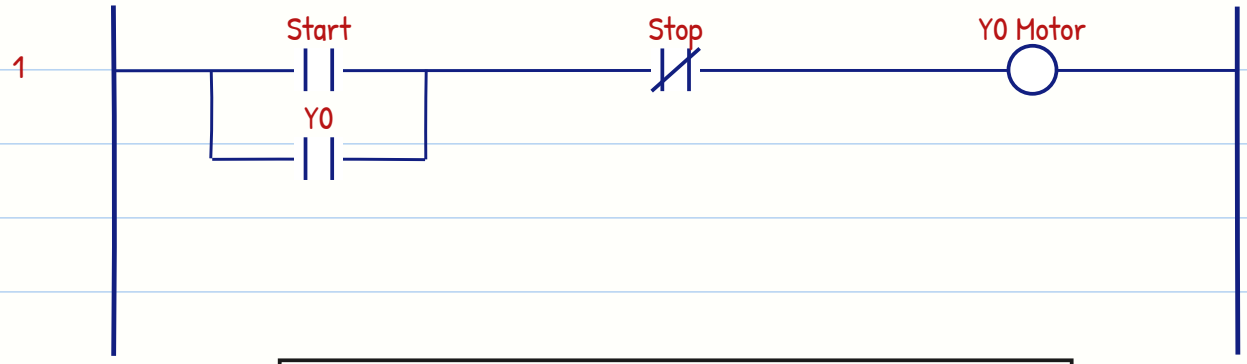


Fig: Latching circuit - motor start / stop

- **SET / RESET (latch / unlatch) instructions:** SET coil makes the output ON and it stays ON even if the input goes OFF until RESET coil clears it. Priority: the last executed instruction wins (reset dominant is safer).

6.3 Interlocking

- Prevent two outputs from being ON together (eg forward and reverse contactors): NC contact of the other output is put in series in each rung.

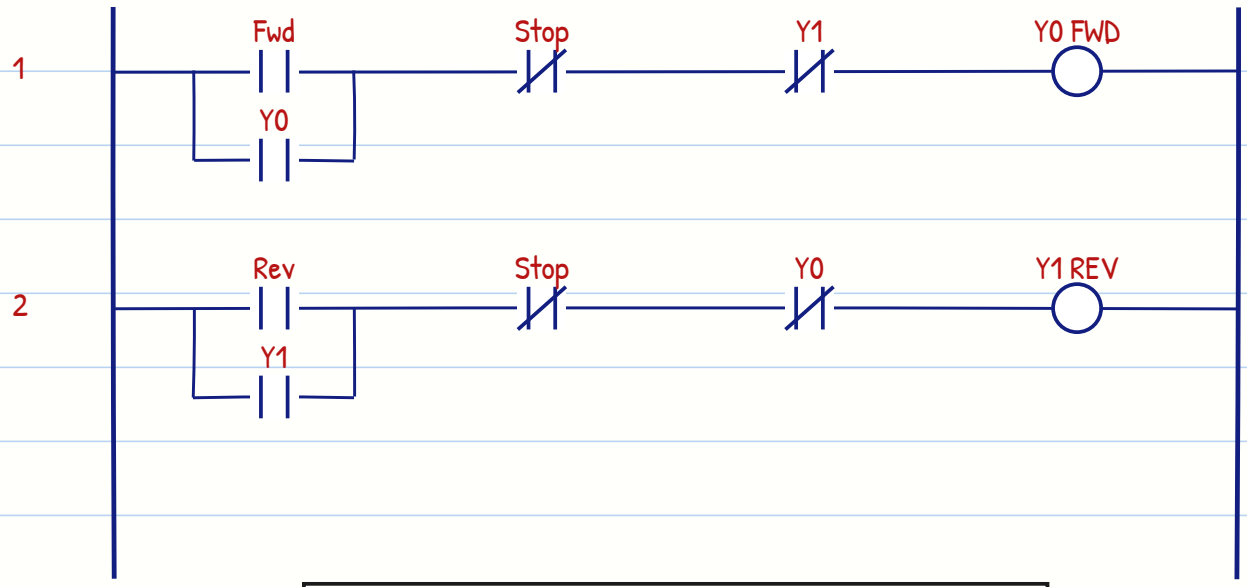


Fig: Forward / Reverse interlock circuit

6.4 Sequencing

- Operations done in a fixed order: step 1 then 2 then 3. Done by using **output of previous step (or its memory bit) as a condition for the next** and the end signal (limit switch, timer) to move on, and the next step **resets** the previous step.
- Eg: Cylinder sequence A+ B+ A- B-: Step 1 A+ ON when Start; limit switch a1 -> Step 2 B+; b1 -> Step 3 A-; a0 -> Step 4 B-; b0 -> end/ready for next cycle. Each step is latched; the next one unlatches the earlier one. Can also use **step counters / shift register / SFC** .

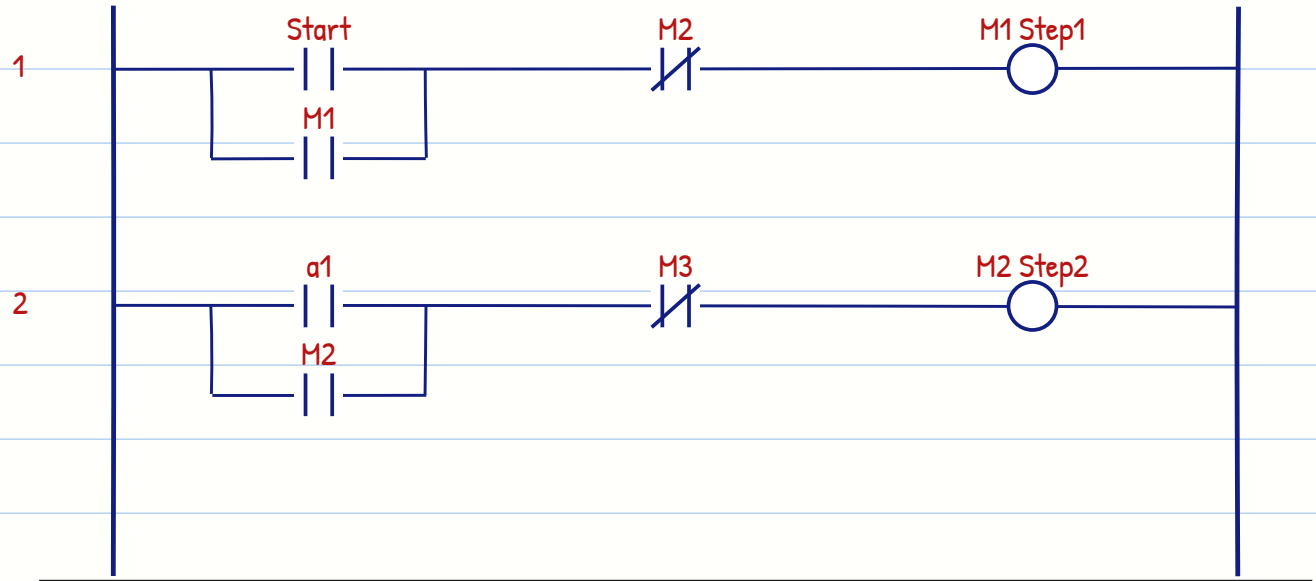


Fig: Sequencing - step 1 enables step 2 which then resets step 1 (M2 interlocks)

7. Timers

- Timers give a time delay. Each has a **preset value (PT), accumulated value (ET), enable input and output bit (Q/DN).** Time base 1 ms, 10 ms, 100 ms or 1 s.
- **TON (On-delay timer):** output turns ON after the preset time **once the input stays ON** ; when the input goes OFF the output resets

immediately. Used for delayed start, motor start sequencing.

- **TOF (Off-delay timer):** output ON immediately with the input; stays ON for the preset time after the input goes OFF. Used for cooling fans and lights.
- **TP (pulse timer):** output pulse of fixed length. **RTO (Retentive timer)** keeps accumulated time on loss of input; needs a reset.

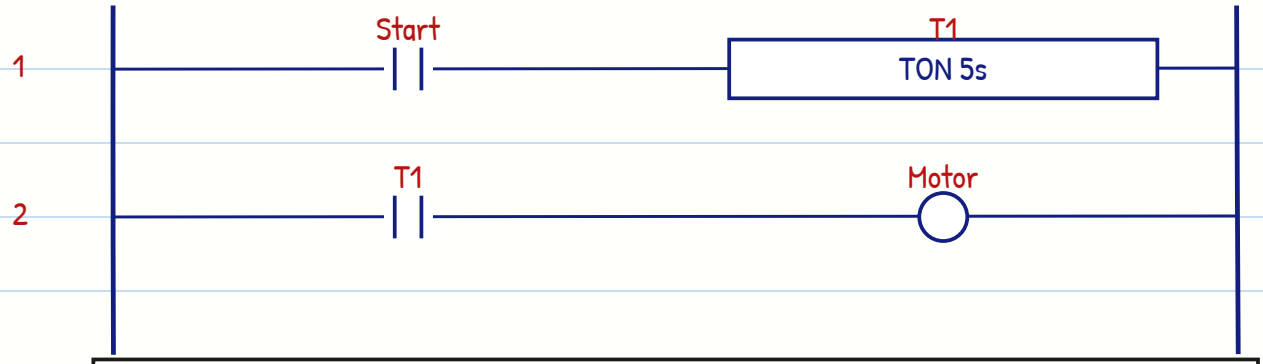


Fig: ON delay timer - Motor starts 5 s after Start is pressed

Timing diagram of TON

- Input goes high at t0 -> timer counts -> at t0 + PT output goes high -> input low -> output low immediately.

Cyclic (flasher) circuit

- Two timers: T1 (on time) and T2 (off time) connected so each one resets the other - lamp blinks. Used in alarms.

8. Relays (control relays) in PLC

- **Internal (auxiliary) relays (M bits / flags)** are software relays - no physical existence - used for storing intermediate logic results, latching, and sequencing.
- **Master control relay (MCR)** switches OFF all the rungs in a zone together (emergency stop). **Jump / Label, subroutine** instructions skip

parts of program.

- **External relays and contactors** are driven by PLC outputs to isolate and switch heavy loads, eg a 24 V DC output driving a 230 V contactor coil.

9. Counters

- Count **events** - parts on a conveyor, bottles, machine cycles. Have **preset value (PV), current value (CV), count input, reset input, and output (Q)** . Counts on the **rising edge** of the input.
- **CTU (Count Up):** CV increases by 1 for each pulse; output ON when CV >= PV. **CTD (Count Down):** CV decreases from PV; output ON when CV = 0. **CTUD (Up-Down counter)** - both.
- **RESET** clears the count to 0 (or loads PV for CTD). Counters are retentive (keep count on power loss in many PLCs).
- **Cascading counters** (Counter output used as the count pulse of the next) for larger counts. **Timer + counter** also gives long time delays.

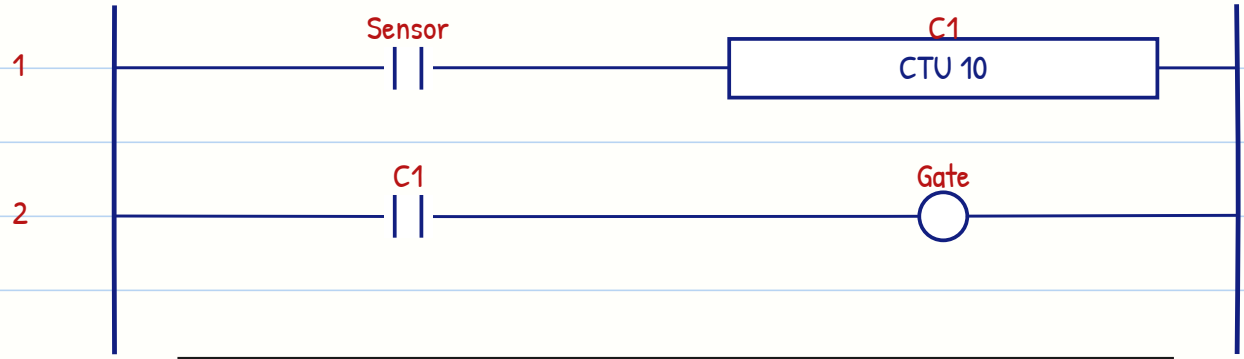


Fig: Counter - gate operates after counting 10 parts

- Eg: Counting 10 bottles on a conveyor: photoelectric sensor counts the bottles -> after 10 the conveyor stops and the packing cylinder operates; the counter is reset.

10. Data Handling in PLC

- PLCs also handle data (not only ON / OFF bits): **bit (1), byte (8), word (16 bit), double word (32 bit)** and floating point (REAL).
- **Data transfer / move instructions:** MOV, copy - load a constant or value of a register to another.
- **Arithmetic instructions:** ADD, SUB, MUL, DIV, increment / decrement, square root.
- **Comparison instructions:** equal (EQ), not equal, greater than (GT), less than (LT), GE, LE - used in rungs as contacts, eg: temperature > 80 -> fan ON.
- **Logical and bit operations:** AND, OR, XOR, NOT on words, shift and rotate (shift register), conversion (BCD to binary).
- **Data registers (D, DB):** store variable values such as set points, preset values, recipes, production counts. **Table / array and sequencer instructions.**
- **Data communication:** with HMI, SCADA, drives and other PLCs by Ethernet / Profibus / Modbus / RS485.

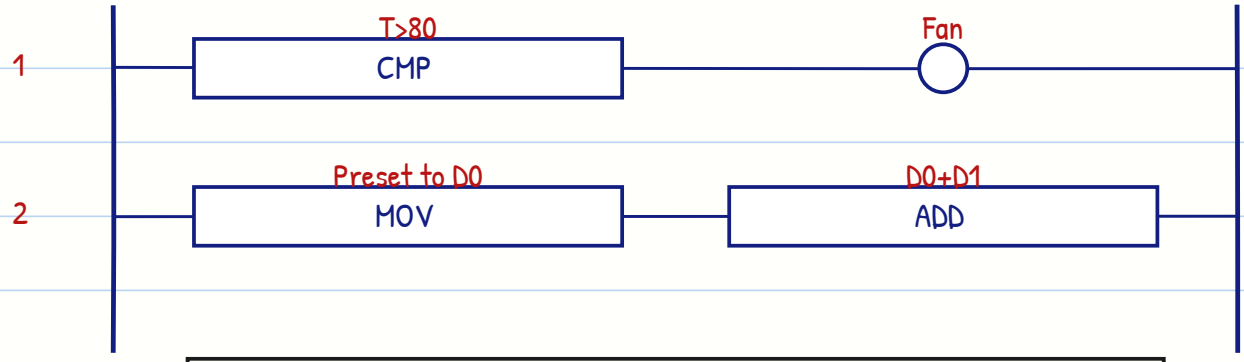


Fig: Data handling - compare and arithmetic rungs

II. Analog Input / Output

- Real world quantities (temperature, pressure, level, speed) are **analog (continuous)** . A PLC works with digital numbers, so **analog modules** are used.

Analog Input (AI) module

- Gets signals **0-10 V, +/-10 V, 0-20 mA or 4-20 mA** from transmitters / sensors (RTD, thermocouple modules). An **ADC (Analog to Digital Converter)** converts it to a digital number (12 bit: 0 - 4095, 16 bit: 0-65535 or 0 - 27648 in Siemens). Resolution = Full scale / 2^n.
- **4-20 mA** is preferred - low noise, long distance, and 4 mA (live zero) shows a wire break.
- **Scaling:** Engineering value = (Count / Max count) x (EUmax - EUmin) + EUmin. Eg: 0-10 bar sensor on 12 bit ADC: count 2048 = 5 bar.

Analog Output (AO) module

- **DAC (Digital to Analog Converter)** changes a digital number to 0-10 V or 4-20 mA to drive **VFD speed reference, proportional valve, control valve positioner, servo.**

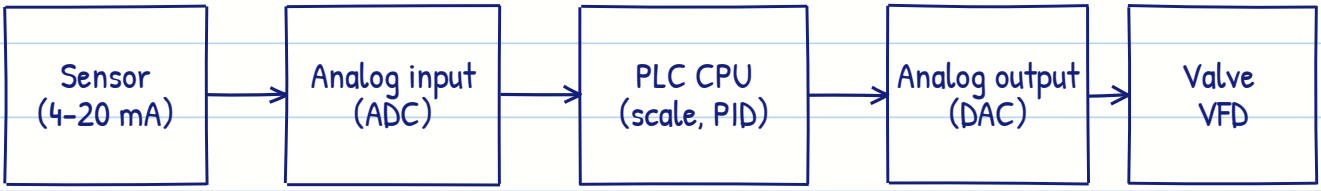


Fig: Analog signal path in a PLC (PID control loop)

- **PID function block:** CPU compares the set point and the process value (analog input) and sends the control output (analog output) - temperature / pressure / flow loops.

12. Selection of Controllers

- **Number and type of I/O** (digital, analog, special - high speed counter, pulse train for stepper/servo) and **20 - 30% spare I/O** for expansion.
- **Speed (scan time) and processing power;** memory size for program and data.
- **Complexity of the control task** (simple logic, PID, motion, data handling); **programming languages** supported.
- **Environment:** temperature, humidity, vibration, EMI, hazardous place, protection class (IP).
- **Communication** (Ethernet, Profibus, Modbus, OPC), HMI/SCADA support.
- **Cost, availability of spares and service,** familiarity of the staff, software and training cost; vendor support.
- **Reliability, redundancy and safety (SIL) requirement;** power supply; size.

Point	Microprocessor	Microcontroller	PLC
Use	General computing	Embedded product	Industrial control
Environment	Office / lab	Product inside	Harsh factory floor
Programming	Assembly, C, high level	Assembly, C, Embedded C	Ladder, FBD, ST

I/O	External chips	Built in, small	Modular, rugged, isolated
Cost	High for system	Lowest	High
Best for	PC, complex tasks	Mass produced, small units	Automation of machines / plants

Important Questions (Unit IV)

- Draw and explain the architecture of 8085 microprocessor / 8051 microcontroller.
- Compare microprocessor and microcontroller.
- Explain the architecture of a PLC with a block diagram. Write the scan cycle.
- Write ladder diagrams for AND, OR, latching and sequencing (A+ B+ A- B-).
- Explain timers (TON, TOF) and counters (CTU, CTD) in a PLC with examples.
- Explain data handling and analog input / output in PLC. Write the selection criteria for controllers.

MECHATRONICS AND MEMS

Subject Code : 23A03702d

UNIT V : Micro Electro Mechanical Systems (MEMS)

Syllabus: History, Effect of scaling, Fabrication Techniques: Oxidation, Physical Vapor Deposition, Chemical Vapor Deposition, Lithography, Etching, Wafer bonding, LIGA, DRIE, Applications: Lab on chip.

I. Introduction to MEMS

- **MEMS (Micro Electro Mechanical Systems)** are miniature devices (size 1 micron to 1 mm) that integrate **mechanical elements (structures), sensors, actuators and electronics on a common silicon substrate** using micro fabrication technology. Typical feature size: 1 - 100 microns; the whole device a few mm.
- Called **MST (Microsystems Technology)** in Europe and **Micromachines** in Japan.
- Mechanical parts (beams, diaphragms, gears, cantilevers) are made by **micromachining** (selective etching and depositing of layers), electronics by standard **IC (integrated circuit) processes** .
- **Components of a MEMS device:** microsensor, microactuator, microstructure (mechanical) and microelectronics (signal processing).

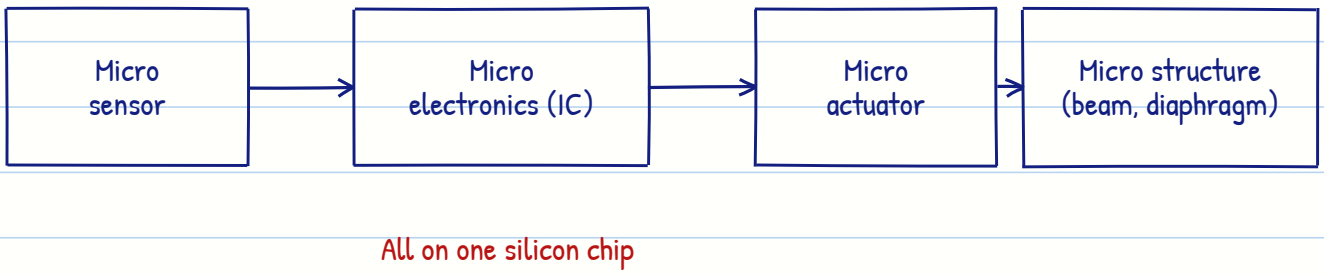


Fig: Elements of a MEMS device

Why silicon?

- High strength to weight, **Young's modulus 190 GPa (like steel)** , high hardness, **no mechanical hysteresis** , good thermal conductivity, **piezoresistive property** , mature IC technology, cheap single crystal wafers, can be bulk and surface micromachined. Other: SiO2, Si3N4, polysilicon, quartz, SiC, polymers (PDMS, SU-8), glass, metals (Au, Al, Ni).

Advantages

- Small size and weight, **low cost in batch fabrication** , low power, high speed and sensitivity, integration with electronics, high reliability, new functions.

Limitations

- High initial cost of the clean room / equipment, packaging is difficult and costly, limited design standard, small forces, **stiction** (sticking of surfaces), difficult testing and calibration.

2. History of MEMS

Year	Event
1947	Invention of transistor (Bell Labs) - base of silicon technology
1954	Piezoresistive effect in silicon and germanium discovered by C.S. Smith
1958	First integrated circuit (Jack Kilby)
1959	Richard Feynman lecture "There's plenty of room at the bottom" - idea of micro machines
1960s	Silicon strain gauges and pressure sensors
1967	Resonant gate transistor (Harvey Nathanson, Westinghouse) - first surface micromachined device
1970s	Bulk micromachined silicon pressure sensors (Kulite, National Semiconductor), anisotropic etching of silicon
1979	Inkjet printer nozzle (HP), micro gas chromatograph (Stanford)

1982	Kurt Petersen paper "Silicon as a mechanical material", LIGA process developed (Germany)
1987-88	Polysilicon micro motors, gears at UC Berkeley, term MEMS used
1990s	ADXL accelerometer (Analog Devices) for air bags, Digital Micromirror Device (Texas Instruments)
2000s	MEMS gyroscopes, microphones, RF MEMS, BioMEMS, lab on chip, MEMS in mobile phones
2010s on	Wearables, IoT, nanoelectromechanical systems (NEMS), MEMS microphones, sensor fusion in smart phones

3. Effect of Scaling

- **Scaling** studies how the physical quantities change when the size of a system is reduced by a scale factor **S** (a length L is reduced to S x L). The **volume** goes as S^3, **area** as S^2, and length as S.
- Different laws of physics dominate at the micro scale - **surface effects dominate volume effects.**

Geometric scaling

- Area ~ L^2 (S^2), Volume ~ L^3 (S^3). **Surface area / Volume ratio ~ 1 / L** - increases as size reduces. Eg: for a cube of L = 1 m ratio = 6, for L = 1 micron ratio is 6 x 10^6.
- So **mass, inertia and weight (~ L^3) fall faster than surface forces (~ L^2)** like friction, adhesion, surface tension, heat transfer, viscosity.

Scaling in different domains

Quantity	Scaling	Meaning at micro scale
Mass / weight / inertia	L^3	Becomes negligible, gravity is unimportant
Surface area	L^2	Surface forces dominate

Friction and adhesion (stiction)	L^2 (compared to L^3)	Parts stick, friction more important than inertia
Electrostatic force	L^2 (F ~ eA V^2 / 2d^2)	Strong, so electrostatic actuation is used
Magnetic force	L^4 (or L^3)	Weak - magnetic actuators are not suitable
Heat capacity	L^3	Very fast heating / cooling
Heat loss (conduction)	L (area/length)	Large heat loss compared to heat stored
Natural frequency	1/L	Very high frequency (MHz), fast response
Stiffness	L	Spring constant falls with L
Power density	1/L	Higher power per unit volume

- **Time scaling:** time for motion t ~ L, so micro devices respond very quickly. **Acceleration** a ~ 1/L (a = 2 s / t^2), **velocity** ~ L.
- **Fluid effects:** viscous forces dominate, flow is **laminar (very low Reynolds number)** - mixing only by diffusion. Surface tension and capillary forces are strong - used in lab on chip.
- **Electrostatic vs magnetic actuation:** electrostatic force scales as L^2, magnetic as L^3 to L^4, so for micro devices **electrostatic (comb drives), piezoelectric and thermal** actuators are used rather than electromagnetic.

Importance of scaling

- Helps to decide which physical principle is best for a micro device; shows **why a design cannot be just a miniature copy** of the macro design.

4. MEMS Fabrication Techniques

- Made with the same steps as the IC process on a **silicon wafer** (single crystal, 100 - 300 mm diameter, 0.5 mm thick) in a **clean room** (class 100 or better).

The main steps: **deposition of thin films, patterning by lithography, and etching** - repeated for each layer.

- Three families: **Bulk micromachining** (etching into the wafer), **Surface micromachining** (depositing and etching thin layers on the wafer surface - using a **sacrificial layer**), and **LIGA / high aspect ratio** processes.

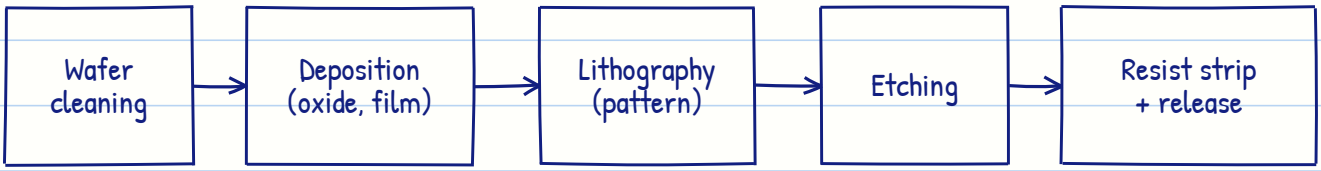


Fig: Basic MEMS process flow

4.1 Oxidation

- Growing a layer of **silicon dioxide (SiO2)** on the silicon wafer by heating the wafer in oxygen or steam at 900-1200 deg C in a furnace. Silicon reacts with the oxidant.
- **Dry oxidation:** Si + O2 -> SiO2. Slow, **dense, high quality, thin oxide** (gate oxide). **Wet oxidation:** Si + 2 H2O -> SiO2 + 2 H2. Faster, thicker (0.5 - 2 micron), lower quality - used for masking and isolation.
- Oxide thickness is given by the **Deal-Grove model:** $x^2 + A x = B (t + tau)$. Initially linear ($x \sim t$), later parabolic ($x \sim \sqrt{t}$). About **46% of the oxide thickness consumes silicon** .
- **Uses:** masking layer for etching and diffusion, electrical insulation, **sacrificial layer** in surface micromachining, passivation, dielectric of capacitors.

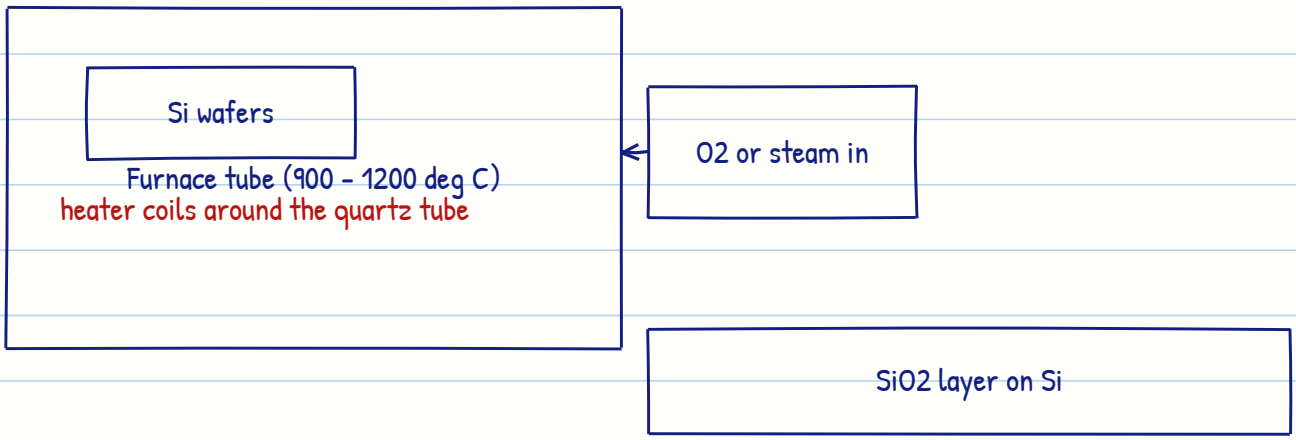


Fig: Thermal oxidation furnace

4.2 Physical Vapor Deposition (PVD)

- A **physical process** in which a solid source material is converted to vapour in vacuum and condenses as a thin film on the wafer. No chemical reaction. Film thickness from nanometres to microns. Used for **metals** (Al, Au, Cr, Ti, Pt).

(a) Evaporation

- The source is heated (**resistive or electron beam heating**) in a high vacuum (10^-6 torr) till it evaporates, and the vapour travels in straight line to the wafer (line of sight) - poor step coverage. Simple and fast.

(b) Sputtering

- A plasma of an inert gas (**Argon ions**) is created in the vacuum chamber. The ions are accelerated to the **target (source material, cathode)** ; by momentum transfer atoms are knocked out (sputtered) and deposit on the wafer (substrate). **DC sputtering** for metals, **RF sputtering** for insulators. **Magnetron sputtering** gives higher rate.
- **Advantages:** good adhesion, **good step coverage** , can deposit alloys and compounds with the same composition as the target, uniform films, low temperature.

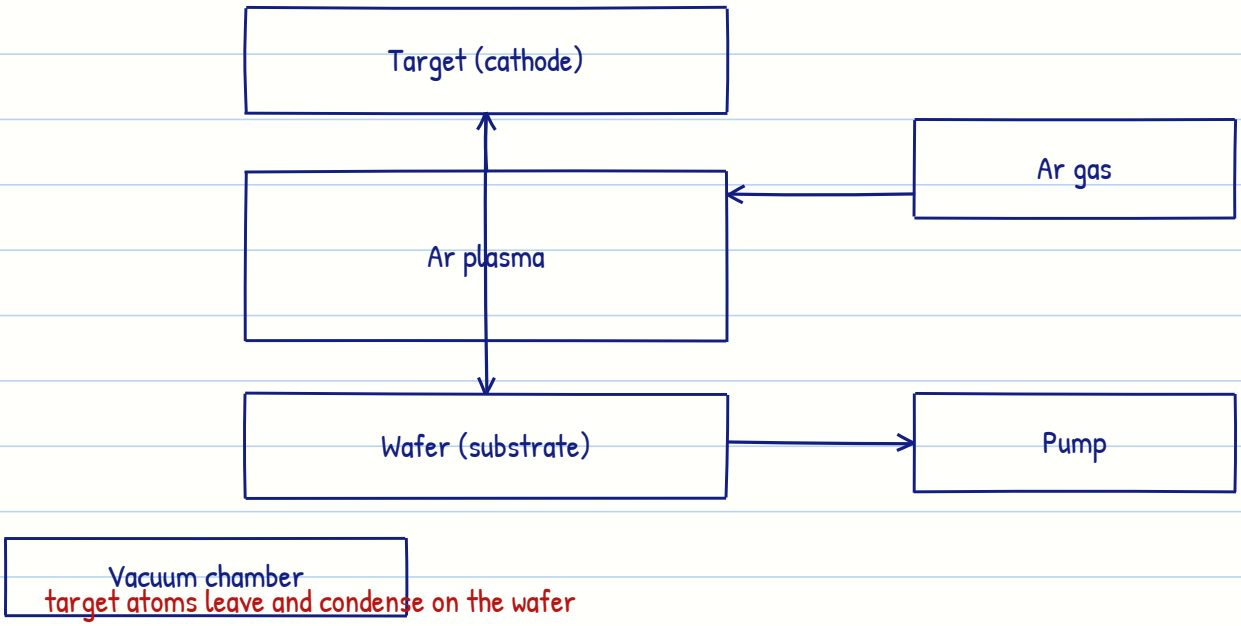


Fig: Sputtering process

4.3 Chemical Vapor Deposition (CVD)

- Thin film is deposited by a **chemical reaction of gases (precursors)** on the heated wafer surface. The by products are removed by the vacuum pump. Gives **dielectrics (SiO2, Si3N4), polysilicon, tungsten** and other films of good uniformity.
- **Steps:** reactant gas transported to the surface -> adsorption -> chemical reaction on the hot surface (film forms) -> by-product desorbs and is pumped out.
- Examples: **Polysilicon:** $SiH_4 \rightarrow Si + 2\ H_2$ (about 600 deg C). **Silicon nitride:** $3\ SiH_2Cl_2 + 4\ NH_3 \rightarrow Si_3N_4 + 6\ HCl + 6\ H_2$ (LPCVD, 800 deg C). **Silicon dioxide:** $SiH_4 + O_2 \rightarrow SiO_2 + 2\ H_2$ (at 400 deg C).

Types

- **APCVD** (atmospheric pressure) - simple, high rate, poor uniformity. **LPCVD** (low pressure, 0.1 - 1 torr) - **best uniformity and step coverage** , high temperature. **PECVD** (plasma enhanced) - **low temperature (200 - 350 deg C)** , can deposit on metals. Also **MOCVD** and **ALD** (atomic layer deposition).

Point	PVD	CVD
Process	Physical (evaporation / sputter)	Chemical reaction of gases
Material	Mostly metals	Dielectrics, poly-Si, metals
Temperature	Low	High (LPCVD), low (PECVD)
Step coverage	Poor (evap.) / good (sputter)	Very good, conformal
Vacuum	High vacuum	Low / atmospheric pressure
Safety	Safer	Toxic, flammable gases

4.4 Lithography

- **Lithography** transfers a **pattern** from a mask to the wafer. It is the most important step - decides the minimum feature size. **Photolithography** uses light and a light sensitive polymer called **photoresist** .

Steps of photolithography

- **1. Cleaning and dehydration bake** of the wafer. **2. Adhesion promoter (HMDS).**
- **3. Spin coating:** a few ml of photoresist is dropped on the wafer and spun at 1000 - 6000 rpm to get a uniform film (0.5 - 2 micron).
- **4. Soft bake** (90-100 deg C) to remove solvent.
- **5. Alignment and exposure:** the **mask (photomask - glass with chrome pattern)** is aligned and the wafer is exposed to UV light (365 nm i-line, 248 nm KrF, 193 nm ArF, EUV 13.5 nm). Types: contact, proximity and projection printing.
- **6. Development:** the exposed (positive resist) or the unexposed (negative resist) areas are dissolved in a developer solution.
- **7. Hard bake** to harden the resist -> then etching or deposition -> **8. Resist stripping** (acetone, O2 plasma).

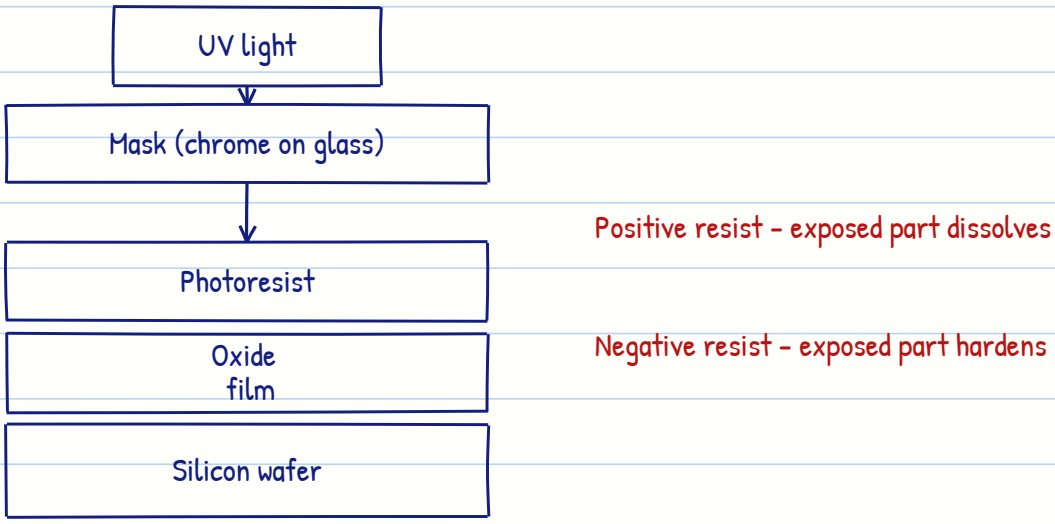


Fig: Photolithography

Positive vs negative resist

- **Positive:** the exposed region becomes soluble - the pattern is same as the mask; better resolution. **Negative:** exposed region becomes insoluble - pattern is the inverse of the mask; good adhesion, lower resolution; eg **SU-8** (thick negative resist for MEMS, LIGA like structures).
- **Other lithography:** electron beam (direct write, 10 nm, slow), X-ray, nanoimprint, **soft lithography (PDMS stamp).**

4.5 Etching

- **Etching** is the selective removal of material not protected by the mask, to create the structures (holes, grooves, membranes). Measures: **etch rate, selectivity** (ratio of etch rate of the material to the mask), **anisotropy** (directionality) and **aspect ratio** .

Wet etching

- Wafer dipped in liquid chemical etchants.
- **Isotropic etching:** equal rate in all directions - undercut below mask; **HNA (HF**

+ HNO3 + acetic acid) for silicon; **HF / buffered HF** for SiO2.

- **Anisotropic etching:** depends on the **crystal plane - KOH, TMAH, EDP** etch Si (100) plane fast, but the (111) plane very slowly (100x less). It forms **V-grooves and pyramidal pits with 54.74 deg walls** ; with (110) wafers gives vertical walls. **Etch stop** by heavy boron doping (p++) or an electrochemical etch stop.
- **Advantages:** simple, low cost, high selectivity, high rate. **Limitations:** limited shapes (crystal planes), undercutting, surface tension problems, not suitable for very small features.

Dry etching

- Uses **gases / plasma** in a vacuum chamber.
- **Plasma etching** (chemical, isotropic), **Reactive Ion Etching (RIE)** - plasma gives reactive ions (CF4, SF6) accelerated vertically toward the wafer: **physical bombardment + chemical reaction -> anisotropic** . **Sputter etching / ion milling** (purely physical, low selectivity). **Vapour phase etching** (XeF2 for Si, HF vapour for SiO2) for releasing structures without stiction.
- **Advantages:** anisotropic, vertical walls, any crystal orientation, high resolution. **Limitations:** expensive equipment, lower selectivity, slower.

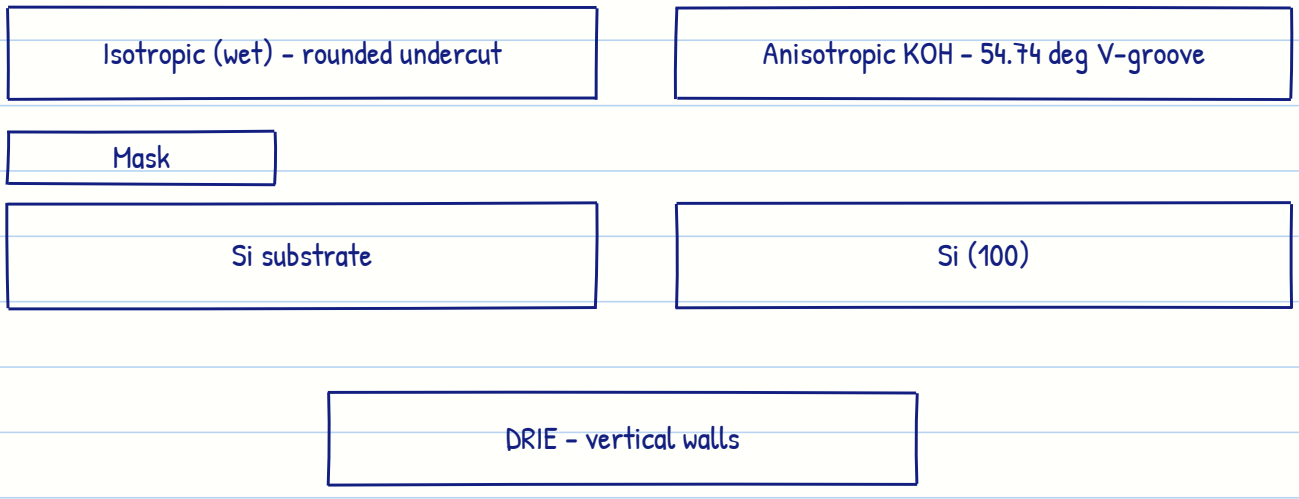


Fig: Types of etch profiles

Bulk and surface micromachining

- **Bulk micromachining:** removes silicon from the wafer by wet / dry etching to form diaphragms, cantilevers and cavities (pressure sensor, accelerometer). Wafer thickness is used.
- **Surface micromachining:** thin films are built on the surface: **sacrificial layer (SiO2 / PSG)** is deposited, then **structural layer (polysilicon)** is deposited and patterned, then the sacrificial layer is **etched away (release)** with HF to free the moving structure (comb drives, gears, cantilever). Care for stiction - use **critical point drying** .

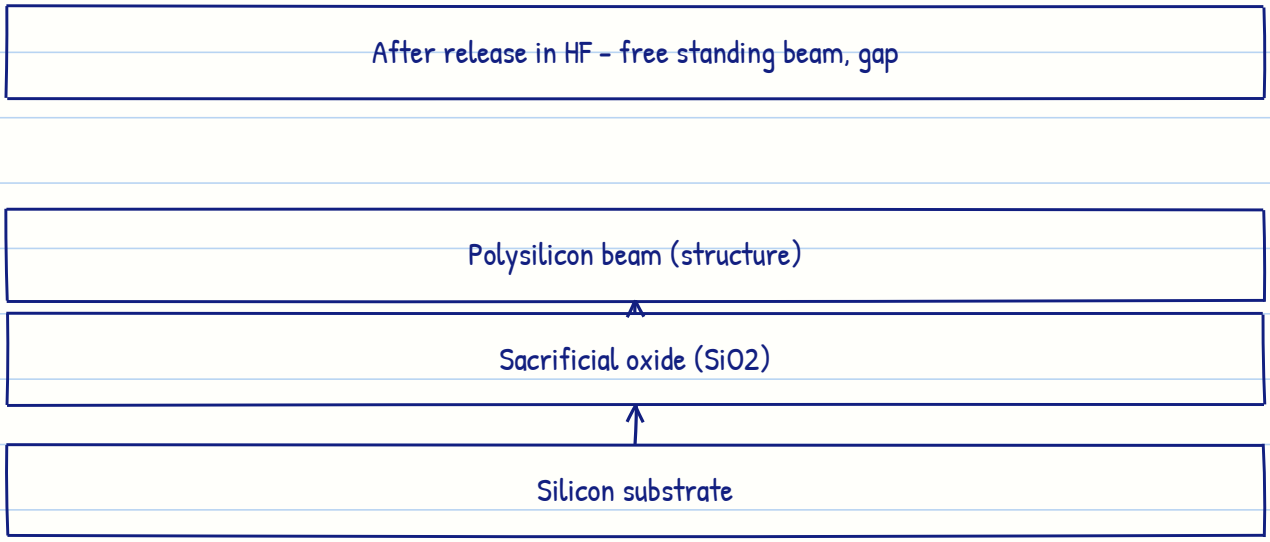


Fig: Surface micromachining - deposit, pattern, and release of a polysilicon cantilever

4.6 DRIE (Deep Reactive Ion Etching)

- A highly anisotropic dry etching to get **deep (100s of microns), vertical walls with a high aspect ratio (> 20:1, up to 100:1)** in silicon. **Bosch process (Robert Bosch, 1990s)** - alternate cycles of:
  - **Etch step:** SF6 plasma isotropically etches silicon a little (few microns).
  - **Passivation step:** C4F8 plasma deposits a **fluorocarbon polymer (Teflon like)** layer on all surfaces.
  - Next etch step: ions bombard the bottom and **remove polymer from the**

floor only (not from the side walls), so the side walls are protected and the etching continues downward. Cycle repeats (**scalloped walls**).

- **Cryogenic process:** wafer cooled to -110 deg C with SF6 + O2 for smooth walls.
- **Selectivity** to oxide mask 200:1 and to photoresist 75:1. Etch rate 1 - 20 micron / min.
- **Uses:** through wafer vias, deep trenches, MEMS accelerometers, gyroscopes, microfluidic channels, comb drives, SOI wafers (oxide as etch stop).

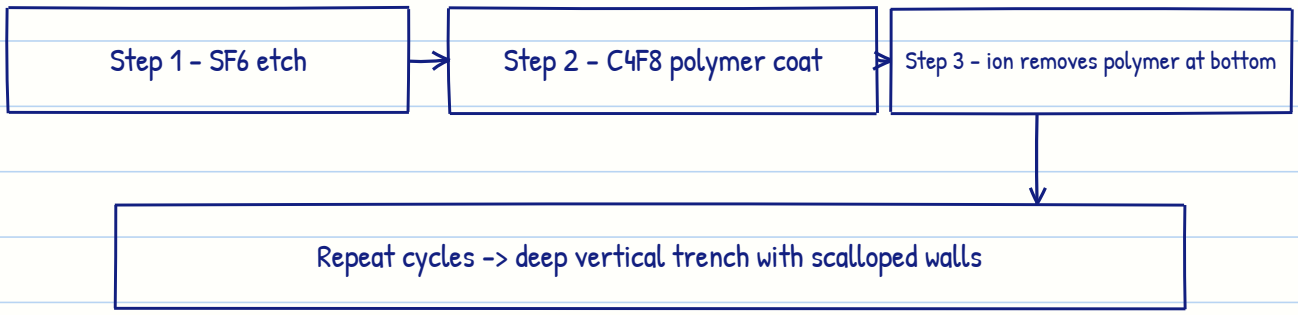


Fig: Bosch process cycle (DRIE)

4.7 Wafer Bonding

- Joining two wafers permanently to make **sealed cavities, multilayer structures, and for packaging** (capping of MEMS), where the single wafer micromachining is not enough.
- **Direct (fusion) bonding:** two very clean, flat, hydrophilic Si / SiO2 wafers are pressed (room temp.), then **annealed at 800 - 1100 deg C** - strong covalent Si-O-Si bonds. Used for **SOI wafers** .
- **Anodic (electrostatic) bonding:** silicon + **Pyrex (borosilicate) glass** heated to **300 - 450 deg C** and a **voltage of 500 - 1000 V** is applied (Si +ve, glass -ve). Na+ ions in glass drift away, creating an electrostatic field and an oxide bond at the interface. Hermetic seal - used for pressure sensors and microfluidics.

- **Intermediate layer bonding: eutectic** (Au-Si, Au-Sn), **glass frit** , **thermo compression** , adhesive (polymer, SU-8, BCB) - lower temperature, bond even with metal / CMOS present.
- **Requirements:** flat, clean, smooth surfaces (roughness < 1 nm), particle free, void free bond; **thermal expansion match** to avoid stress.

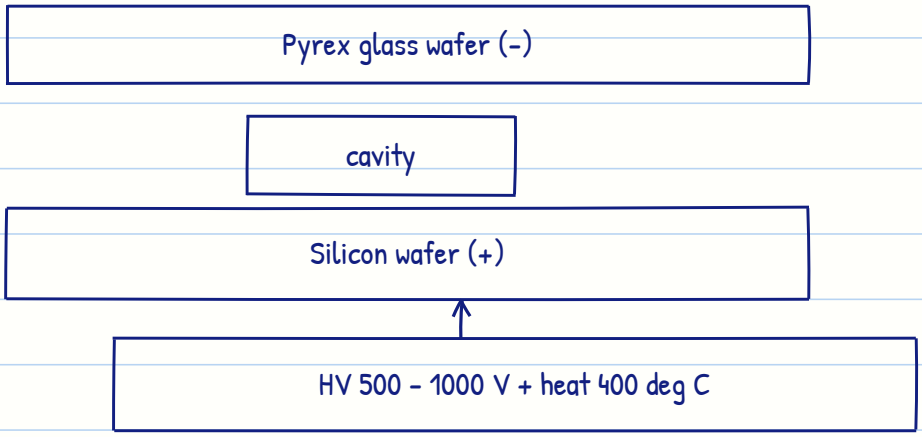


Fig: Anodic bonding of silicon to glass

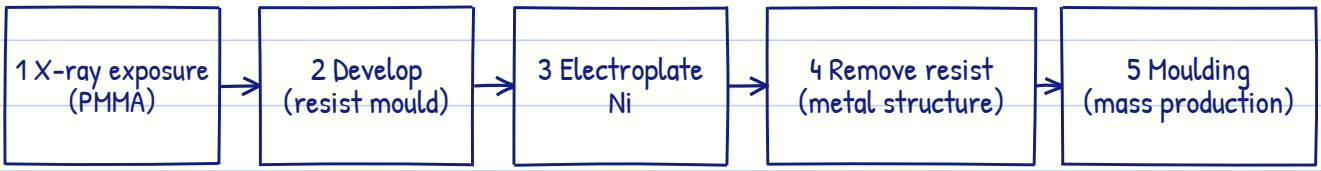
4.8 LIGA Process

- **LIGA** - German acronym: **LI** thographie (lithography), **G** alvanoformung (electroplating) and **A** bformung (moulding). Developed at Karlsruhe, Germany in the early 1980s (Becker and Ehrfeld). Makes **high aspect ratio (up to 100+), 3D metal / plastic / ceramic microstructures** with heights from tens microns to several mm, not limited to silicon.

Steps

- **1. X-ray lithography:** a thick layer (up to 1 mm) of **PMMA (polymethyl methacrylate)** resist on a conductive substrate is exposed through an X-ray mask using **synchrotron X-ray** radiation. The exposed PMMA gets weaker (chain scission).
- **2. Development:** the exposed PMMA is removed - a deep resist mould with very vertical walls.

- **3. Electroplating (Galvanoformung):** metal (**nickel, copper, gold**) is electro deposited into the mould cavities.
- **4. Resist removal:** the PMMA is dissolved - a free standing **metal microstructure** , or used as a **mould insert (mould tool)** .
- **5. Moulding (Abformung):** the metal mould insert is used for **mass production by injection moulding / hot embossing / casting** of plastic parts.



Synchrotron X-ray -> deep and vertical walls

Fig: LIGA process

Variants

- **UV-LIGA / SU-8:** uses UV light with SU-8 resist (cheaper, no synchrotron, height to 1 mm) and **sacrificial LIGA** .
- **Advantages:** very high aspect ratio, vertical smooth walls (roughness < 50 nm), wide range of materials, mass production by moulding. **Limitations:** very **costly** (synchrotron source, X-ray mask), long process, few facilities, mostly not compatible with IC electronics.
- **Applications:** micro gears, micro motors, microfluidic channels, accelerometers, optical switches, fiber optic connectors, micro nozzles, spinnerets.

4.9 Comparison of the micromachining methods

Point	Bulk	Surface	LIGA

Material	Silicon wafer	Thin films on wafer	Metal, plastic, ceramic
Structure size	Hundreds of microns thick	Few microns thick	Up to mm tall
Aspect ratio	Medium (DRIE high)	Low	Very high
Cost	Medium	Medium	High
IC compatibility	Good	Very good	Poor
Eg	Pressure sensor, nozzle	Comb drive, gyroscope	Micro gears, moulds

5. Applications of MEMS

- **Automotive:** air bag accelerometer (ADXL), tyre pressure (TPMS), manifold absolute pressure (MAP), gyroscope for ESC, fuel injection nozzle.
- **Consumer / mobile:** accelerometer and gyroscope (screen rotation, games), MEMS microphone, compass (magnetometer), inkjet print head, **DMD (digital micromirror device) projectors** .
- **Medical (BioMEMS):** disposable blood pressure sensor, pacemaker sensor, micro needles for drug delivery, lab on chip, DNA chips, hearing aid, endoscope.
- **Aerospace / defence:** inertial navigation (gyros), micro satellites, RF MEMS switches, drones.
- **Industrial:** vibration, pressure, flow and gas sensors, optical MEMS switches (fibre optic communication), micro mirrors, **RF filters** , energy harvesters.

6. Lab on Chip (LOC)

- A **lab on a chip** is a device of a few mm to cm size that integrates **one or many laboratory functions (sample preparation, mixing, reaction, separation, detection) on a single chip** using **microfluidics** - handling tiny volumes of fluid (nano to picolitre) in channels of 10 - 100 micron width. Also called **micro Total Analysis System (uTAS)** .
- **Principle:** at micro scale the flow is **laminar** (low Reynolds number), surface

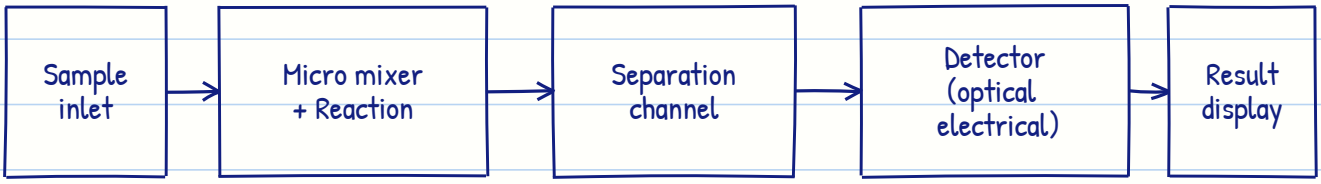
forces dominate, diffusion is fast (short distances) - so mixing is by diffusion; fluid movement is by **pressure, capillary action, electro-osmosis, electrophoresis, electrowetting** or micro pumps.

Main parts

- **Sample inlet / reservoirs, microchannels, micro pumps and valves, micro mixers, reaction chambers, separation unit** (capillary electrophoresis), **detectors** (optical fluorescence, electrochemical), **heaters, electrodes** , and a readout electronics.

Materials and fabrication

- **Glass and silicon** (etching, anodic bonding), and **polymers - PDMS (soft lithography), PMMA, SU-8, PC, COC** (hot embossing, injection moulding, laser cutting). **Paper based** low cost chips. Channels are sealed by bonding a cover plate.



Pumps, valves and electrodes on the same chip

Fig: Lab on chip - functional blocks

Advantages

- **Very small sample and reagent volume,** fast analysis (minutes), low cost per test, **portable and point of care** use, automation and parallel testing, low power and waste, high sensitivity and safe.

Limitations

- Detection at small volumes is difficult, clogging of channels and bubbles, the

cost of the fabrication / packaging, interface with the macro world (tubing), standardisation and reliability issues.

Applications

- **Medical diagnostics** - glucose monitoring, pregnancy test, blood analysis, **COVID / infectious disease tests,** DNA / PCR chips, drug discovery and delivery, **organ on chip** ; environmental monitoring (water testing), food safety, forensic, chemical synthesis (micro reactor), and biological cell sorting and counting.

Important Questions (Unit V)

- What is MEMS? Explain the history of MEMS and the advantages of MEMS.
- Explain the effect of scaling on MEMS devices (geometric, electrostatic, heat, fluid).
- Explain oxidation and the PVD / CVD processes used in MEMS fabrication.
- Explain the photolithography process with a neat sketch.
- Explain wet and dry etching; bulk and surface micromachining; DRIE (Bosch process).
- Explain wafer bonding techniques and the LIGA process.
- Write a short note on Lab on chip and applications of MEMS.